

FEATURES

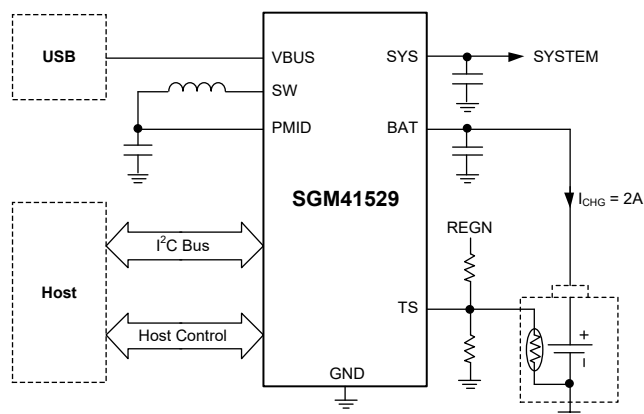
- 3.9V to 6.2V Input Voltage Range with up to 18V Maximum Input Voltage
- High-Efficiency 2A, 1.5MHz Boost Mode Charger:
 - Up to 94% Charge Efficiency with 5V Adaptor, 7.6V Battery, and 1A Charge Current
- USB On-The-Go (OTG) Buck Mode:
 - 4.5V to 5.5V Adjustable Output
 - Up to 2A Output
 - Up to 95% Efficiency at 5.1V/1A Output
- High Charge Voltage and Current Setting Accuracy:
 - $\pm 0.6\%$ Charge Voltage Regulation
 - $\pm 5\%$ Charge Current Regulation
- Selectable PFM Mode and Out-of-Audio (OOA) Mode at Light Load Operations
- USB BC1.2 and Non-Standard Adaptors Auto Detection Based on D+/D- Inputs
- Narrow Voltage DC (NVDC) and Dynamic Power Management:
 - Programmable Input Voltage Limit (VINDPM)
 - Programmable Input Current Limit (IINDPM)
- Input Current Optimizer (ICO) to Maximize Adaptor Output Current without Overloading
- I²C Port for Flexible System Parameter Setting and Status Reporting
- Integrated 16-Bit ADC for Monitoring Input Voltage, Input Current, Battery Voltage, Charge Current, System Voltage, Battery Temperature and Die Temperature
- Fully Integrated All MOSFETs, Current Sensing and Loop Compensation
- High Battery Discharge Efficiency with Low R_{DS(on)} (18m Ω) Switch

- Fully Safety and Protections
 - Input/System/Battery Over-Voltage Protection
 - Output Over-Current Protection for Both Boost and OTG Buck Mode
 - Battery Charging Safety Timer
 - Battery Temperature Sensing in Charge and OTG Mode
 - Thermal Regulation and Thermal Shutdown
- Available in a Green TQFN-4x4-24L Package

APPLICATIONS

Smart Phones, EPOS
 Bluetooth Speaker
 Wireless Security Camera
 Portable Internet Devices and Accessory

SIMPLIFIED SCHEMATIC



GENERAL DESCRIPTION

The SGM41529 is a battery charger and system power path management device for 2-cell Li-Ion or Li-polymer batteries. Its low-impedance power path optimizes efficiency, reduces battery charging time, and extends battery life. I²C programming makes it a flexible powering and charger design solution.

The SGM41529 can detect the input source types which include SDP/CDP/DCP and non-standard adaptor through the D+/D- pins following USB BC1.2 specification.

The SGM41529 features a dynamic power management (DPM) to avoid input adaptor overload or meet the maximum current limit. It keeps the system voltage regulated to its minimum setting in DPM mode by reducing the charge current. The SGM41529 also provides supplement mode to further support system output in case that the charge current decreased to zero and the input is still overloaded. The SGM41529 can provide the maximum power point with an algorithm called input current optimizer (ICO) to avoid the input source overload.

The SGM41529 supports the default mode (standalone) without host control. It automatically detects the battery voltage and starts the charge. If the device is not in thermal regulation or DPM mode, the charge cycle automatically terminates when a full charge is detected. The charger automatically initiates a new charging cycle if the battery voltage falls below the recharge threshold.

The SGM41529 supports USB On-The-Go (OTG) operation by supplying default 5.1V on the VBUS with an output current limit up to 2A.

The SGM41529 provides full protections for safety of battery charging and system operation, including battery temperature monitoring, charging safety timer, over-current and over-voltage protections. When any fault occurs, the SGM41529 asserts a nINT pulse to notify the host.

The SGM41529 is available in a Green TQFN-4×4-24L package.

I²C Controlled 2A, 2-Cell Boost Battery Charger with Power Path, OTG for USB Input

SGM41529

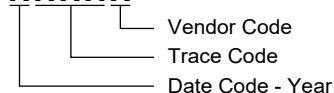
PACKAGE/ORDERING INFORMATION

MODEL	PACKAGE DESCRIPTION	SPECIFIED TEMPERATURE RANGE	ORDERING NUMBER	PACKAGE MARKING	PACKING OPTION
SGM41529	TQFN-4×4-24L	-40°C to +85°C	SGM41529YTQF24G/TR	SGM41529 YTQF24 XXXXXX	Tape and Reel, 3000

MARKING INFORMATION

NOTE: XXXXX = Date Code, Trace Code and Vendor Code.

XXXXX



Green (RoHS & HSF): SG Micro Corp defines "Green" to mean Pb-Free (RoHS compatible) and free of halogen substances. If you have additional comments or questions, please contact your SGMICRO representative directly.

ABSOLUTE MAXIMUM RATINGS

Voltage Range (with Respect to GND)

VBUS (Converter Not Switching)	-0.3V to 18V
PMID (Converter Not Switching)	-0.3V to 8.5V
BAT, SYS (Converter Not Switching)	-0.3V to 12V
SW	-0.3V to 13V
BTST	-0.3V to 19V
REGN, D+, D-, SDA, SCL, TS, STAT, nPG, nCE, nINT	-0.3V to 6V
ILIM	-0.3V to 5V
BTST to SW	-0.3V to 6V
Output Sink Current, nINT, STAT, nPG	6mA (MAX)
Package Thermal Resistance	
TQFN-4×4-24L, θ_{JA}	29.2°C/W
TQFN-4×4-24L, θ_{JB}	7.6°C/W
TQFN-4×4-24L, θ_{JC} (TOP)	23.2°C/W
TQFN-4×4-24L, θ_{JC} (BOT)	2°C/W
Junction Temperature	+150°C
Storage Temperature Range	-65°C to +150°C
Lead Temperature (Soldering, 10s)	+260°C
ESD Susceptibility ^{(1) (2)}	
HBM	±2000V
CDM	±1000V

NOTES:

- For human body model (HBM), all pins comply with ANSI/ESDA/JEDEC JS-001 specifications.
- For charged device model (CDM), all pins comply with ANSI/ESDA/JEDEC JS-002 specifications.

DISCLAIMER

SG Micro Corp reserves the right to make any change in circuit design, or specifications without prior notice.

RECOMMENDED OPERATING CONDITIONS

Input Voltage Range, V_{VBUS}	3.9V to 6.2V
Average Input Current (VBUS), I_{VBUS}	3.3A (MAX)
Average Charge Current (BAT), I_{BAT}	2.2A (MAX)
RMS Discharging Current with Internal MOSFET, I_{BAT_RMS}	5A (MAX)
Peak Discharging Current with Internal MOSFET, I_{BAT_PK}	9A (up to 1μs) (MAX)
Battery Voltage, V_{BAT}	9.2V ⁽³⁾ (MAX)
Operating Ambient Temperature Range	-40°C to +85°C

NOTE:

- The voltage spikes on SW pins should be less than the absolute maximum rating. Following the layout guidelines is helpful to minimize the switching noise.

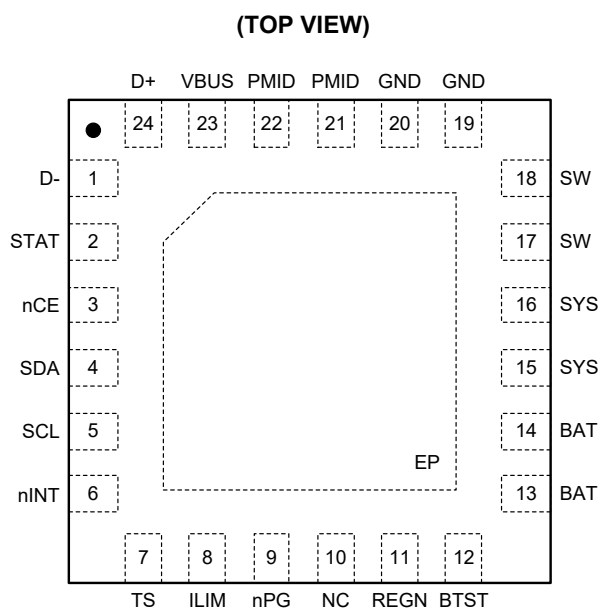
OVERSTRESS CAUTION

Stresses beyond those listed in Absolute Maximum Ratings may cause permanent damage to the device. Exposure to absolute maximum rating conditions for extended periods may affect reliability. Functional operation of the device at any conditions beyond those indicated in the Recommended Operating Conditions section is not implied.

ESD SENSITIVITY CAUTION

This integrated circuit can be damaged if ESD protections are not considered carefully. SGMICRO recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage. ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because even small parametric changes could cause the device not to meet the published specifications.

PIN CONFIGURATION



TQFN-4x4-24L

PIN DESCRIPTION

PIN	NAME	TYPE	FUNCTION
1	D-	AIO	Negative Line of the USB Data Line Pair. D+/D- based USB device protocol detection.
2	STAT	DO	Open-Drain Charge Status Output. Use a 10kΩ pull-up to the logic high rail (or an LED + a resistor). The STAT pin acts as follows: During charging: low (LED ON). Charge completed or charge disabled: high (LED OFF). Charge suspended (in response to a fault): pulses with 1Hz, 50% duty cycle (LED BLINKS). The function can be disabled via STAT_DIS bit.
3	nCE	DI	Charge Enable Input Pin (Active Low). When nCE pin is low and EN_CHG = 1, the charge is enabled. Do not leave this pin floating.
4	SDA	DIO	I ² C Data Signal. Use a 10kΩ pull-up to the logic high rail.
5	SCL	DI	I ² C Clock Signal. Use a 10kΩ pull-up to the logic high rail.
6	nINT	DO	Open-Drain Interrupt Output Pin. Use a 10kΩ pull-up to the logic high rail. The nINT pin sends out a negative 256μs pulse to the host when a fault occurs or a new charge status updates.
7	TS	AI	Temperature Sense Input Pin. Connect to the battery NTC thermistor that is grounded on the other side. To program operating temperature window, it can be biased by a resistor divider between REGN and GND. When TS pin voltage goes to hot or cold range, the charge suspends. It is recommended to use a 103AT-2 type thermistor.
8	ILIM	AI	Input Current Limit. A resistor between ILIM and GND pins can clamp the input current limit as $I_{INMAX} = K_{ILIM}/R_{ILIM}$. When EN_ILIM = 1, the lower limit of IINDPM registers and ILIM pin resistor setting sets the actual input current limit. The ILIM pin can supply higher than 500mA current limit only. If the current limit is not triggered, the ILIM pin voltage indicates the actual input current limit by $I_{IN} = K_{ILIM} \times V_{ILIM}/(R_{ILIM} \times 0.8V)$. Change EN_ILIM = 0 can disable both input current limit clamping and input current monitor function from ILIM pin.

SGM41529 I^2C Controlled 2A, 2-Cell Boost Battery Charger with Power Path, OTG for USB Input

PIN DESCRIPTION (continued)

PIN	NAME	TYPE	FUNCTION
9	nPG	DO	Open-Drain Input Power Good Indicator Pin. Use a 10k Ω pull-up to the logic high rail. If the input voltage is below V_{VBUS_OV} , and passes the poor source detection, this pin outputs a low state to indicate a good input.
10	NC	–	Internal No Connection.
11	REGN	P	Gate Drive Power Supply. Power supply for internal MOSFET drivers and IC. Connect a 4.7 μ F ceramic capacitor from REGN pin to GND. It is recommended to place the capacitor close to REGN pin.
12	BTST	P	PWM High-side Driver Supply. It is internally connected to the bootstrap diode cathode. A 47nF bootstrap capacitor between SW pin and BTST pin is recommended.
13, 14	BAT	P	Battery Positive Terminal Pin. Use a 10 μ F capacitor between BAT and GND pins close to the device. SYS and BAT pins are internally connected by BATFET with current sensing capability.
15, 16	SYS	P	System Connection. The SYS pin is connected to BAT pin with an internal BATFET. It is recommended to connect more than 44 μ F ceramic capacitor between SYS and GND pins as close to the device as possible.
17, 18	SW	P	Switching Node Output. An external inductor is connected between PMID and SW pins.
19, 20	GND	–	Ground Pin of the Device.
21, 22	PMID	P	PMID Pin. Connect to the source of blocking MOSFET (QBLK). Given the total IC internal input supply. Connect a 10 μ F ceramic capacitor from PMID pin to GND.
23	VBUS	P	Charger Input (V_{IN}). Connect VBUS to the external DC power supply. Place a 1 μ F ceramic capacitor from VBUS pin to GND close to the device.
24	D+	AIO	Positive Line of the USB Data Line Pair. D+/D- based USB device protocol detection.
Exposed Pad	EP	–	Exposed Pad. Connect to GND.

NOTE: AI = analog input, AO = analog output, AIO = analog input and output, DI = digital input, DO = digital output, DIO = digital input and output, P = power.

SGM41529

I²C Controlled 2A, 2-Cell Boost Battery Charger with Power Path, OTG for USB Input

ELECTRICAL CHARACTERISTICS

($V_{VBUS_UVLOZ} < V_{VBUS} < V_{VBUS_OV}$, $T_J = -40^{\circ}\text{C}$ to $+85^{\circ}\text{C}$, typical values are at $T_J = +25^{\circ}\text{C}$, unless otherwise noted.)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
Quiescent Currents						
Battery Discharge Current (BAT)	I_{BAT}	$V_{BAT} = 9\text{V}$, no VBUS, SCL, SDA = 0V or 1.8V, ADC disabled	$T_J = +25^{\circ}\text{C}$	13	17	μA
			$T_J < +85^{\circ}\text{C}$	13	21	
Input Supply Current (VBUS) in HIZ	I_{VBUS_HIZ}	$V_{VBUS} = 5\text{V}$, HIZ mode, no battery, ADC disabled	$T_J = +25^{\circ}\text{C}$	20	24	μA
			$T_J < +85^{\circ}\text{C}$	20	29	
Input Supply Current (VBUS)	I_{VBUS}	$V_{VBUS} = 5\text{V}$, $V_{BAT} = 7.6\text{V}$, converter not switching		2.7	3.5	mA
		$V_{VBUS} = 5\text{V}$, $V_{BAT} = 7.6\text{V}$, $I_{SYS} = 0\text{A}$, converter switching, OOA disabled		3.5		
Battery Discharge Current in OTG Mode	I_{BAT_OTG}	$V_{BAT} = 8.4\text{V}$, OTG Buck mode, $I_{VBUS} = 0\text{A}$, converter switching, OOA disabled		2.5		mA
BAT Pin and VBUS Pin Power-Up						
VBUS Operating Range	V_{VBUS_OP}		3.9		6.2	V
VBUS UVLO to Have Active I ² C (with No Battery)	V_{VBUS_UVLOZ}	V_{VBUS} rising		3.3	3.67	V
VBUS Minimum (as One of the Conditions) to Turn on REGN	$V_{VBUS_PRESENT}$	V_{VBUS} rising		3.7	3.9	V
VBUS Over-Voltage Rising Threshold	V_{VBUS_OV}	V_{VBUS} rising, CCM switching	6.2		6.6	V
VBUS Over-Voltage Falling Threshold		V_{VBUS} falling, CCM switching	5.9		6.4	V
BAT Voltage to Have Active I ² C (No Source on VBUS)	V_{BAT_UVLOZ}	V_{BAT} rising	3.1	3.6	4.1	V
Bad Adaptor Detection Threshold	V_{BAD_SRC}	V_{VBUS} falling		3.6		V
Bad Adaptor Detection Current Source	I_{BAD_SRC}			15		mA
Power Path Management						
System Regulation Voltage	V_{SYS}	$I_{SYS} = 0\text{A}$, $V_{BAT} = 8.8\text{V}$, $V_{BAT} > \text{SYS_MIN}[3:0]$, charge disabled (EN_CHG = 0)		$V_{BAT} + 0.2$		V
		$I_{SYS} = 0\text{A}$, $V_{BAT} < \text{SYS_MIN}[3:0]$, charge disabled (EN_CHG = 0)		$V_{SYS_MIN} + 0.3$		
Minimum DC System Voltage Output	V_{SYS_MIN}	$V_{BAT} < \text{SYS_MIN}[3:0] = 0010$, charge disabled (EN_CHG = 0)	6.3	6.5		V
Battery Charger						
Charge Voltage Program Range	V_{REG_RANGE}		6.8		9.2	V
Typical Charge Voltage Step	V_{REG_STEP}			10		mV
Charge Voltage	V_{REG_ACC}	$V_{REG} = 8.4\text{V}$, $T_J = +25^{\circ}\text{C}$	8.35	8.4	8.45	V
Charge Current Regulation Range	I_{CHG_RANGE}		100		2200	mA
Charge Current Regulation Step	I_{CHG_STEP}			50		mA
Fast Charge Current Regulation Accuracy	I_{CHG_ACC}	$V_{BAT} = 6.2\text{V}$ or 7.6V , $T_J = +25^{\circ}\text{C}$	$I_{CHG} = 1000\text{mA}$	-5	5	%
			$I_{CHG} = 500\text{mA}$	-15	15	
Pre-Charge Current Range	I_{PRECHG_RANGE}		50		800	mA
Typical Pre-Charge Current Step	I_{PRECHG_STEP}			50		mA
Pre-Charge Current Accuracy	I_{PRECHG_ACC}	$V_{BAT} = 5.2\text{V}$, $I_{PRECHG} = 200\text{mA}$	$T_J = +25^{\circ}\text{C}$	145	235	mA
			$T_J = 0^{\circ}\text{C}$ to $+85^{\circ}\text{C}$	140	244	
Termination Current Range	I_{TERM_RANGE}		50		800	mA
Typical Termination Current Step	I_{TERM_STEP}			50		mA
Termination Current Accuracy	I_{TERM_ACC}	$I_{CHG} = 1.5\text{A}$, $I_{TERM} = 150\text{mA}$	$T_J = +25^{\circ}\text{C}$	40	230	mA
			$T_J = 0^{\circ}\text{C}$ to $+85^{\circ}\text{C}$	22	257	

I²C Controlled 2A, 2-Cell Boost Battery Charger SGM41529 with Power Path, OTG for USB Input

ELECTRICAL CHARACTERISTICS (continued)

($V_{VBUS_UVLOZ} < V_{VBUS} < V_{VBUS_OV}$, $T_J = -40^{\circ}\text{C}$ to $+85^{\circ}\text{C}$, typical values are at $T_J = +25^{\circ}\text{C}$, unless otherwise noted.)

PARAMETER	SYMBOL	CONDITIONS		MIN	TYP	MAX	UNITS
Battery Short Voltage Rising Threshold to Start Pre-Charging	V _{BAT_SHORT_RISING}	V _{BAT} rising		4.27	4.4	4.49	V
Battery Short Voltage Falling Threshold to Stop Pre-Charging	V _{BAT_SHORT_FALLING}	V _{BAT} falling		3.79	4	4.13	V
Battery Low Voltage Trickle Charging Current	I _{BAT_SHORT}	V _{BAT} < 4.4V			80		mA
VBAT_LOW Rising Threshold to Start Fast-Charging	V _{BAT_LOW_RISING}	V _{BAT} rising	VBATLOW = 1 (6V)	5.8	6	6.2	V
			VBATLOW = 0 (5.6V)	5.4	5.6	5.8	
VBAT_LOW Falling Threshold to Stop Fast-Charging	V _{BAT_LOW_FALLING}	V _{BAT} falling	VBATLOW = 1 (6V)	5.3	5.6	5.8	V
			VBATLOW = 0 (5.6V)	4.9	5.2	5.4	
Recharge Threshold below V _{REG}	V _{RECHG}	V _{BAT} falling	VRECHG[1:0] = 01		200		mV
			VRECHG[1:0] = 10		300		
High-side Switching MOSFET On-Resistance between SW and SYS (Q2)	R _{ON_QHS}	T _J = +25°C			30	35	mΩ
		T _J = -40°C to +85°C			30	43	
Low-side Switching MOSFET On-Resistance between SW and GND (Q3)	R _{ON_QLS}	T _J = +25°C			34.5	40	mΩ
		T _J = -40°C to +85°C			34.5	50	
MOSFET On-Resistance between SYS and BAT (Q4)	R _{ON_QBAT}	T _J = +25°C			18	20.5	mΩ
		T _J = -40°C to +85°C			18	25.2	
BAT Discharge Current Source	I _{BAT_DISCHG}	V _{BAT} = 8V, EN_BAT_DISCHG = 1			11.5		mA
Input Voltage/Current Regulation							
Input Voltage Regulation Range	V _{INDPM_RANGE}			3.9		5.5	V
Input Voltage Regulation Step	V _{INDPM_STEP}				100		mV
Input Voltage Limit	V _{INDPM}	V _{INDPM} = 3.9V		3.825	3.9	3.984	V
		V _{INDPM} = 4.4V		4.325	4.4	4.477	
Input Current Regulation Range	I _{INDPM_RANGE}			500		3300	mA
Input Current Regulation Step	I _{INDPM_STEP}				100		mA
Input Current Regulation Limit	I _{INDPM_ACC}	I _{INDPM} = 500mA		347	458	547	mA
		I _{INDPM} = 900mA		724	833	916	
		I _{INDPM} = 2500mA		2168	2331	2500	
		I _{INDPM} = 3000mA		2610	2804	3000	
Charge Current Setting Ratio	K _{ILIM}	I _{INMAX} = K _{ILIM} /R _{ILIM} , input current regulation by ILIM pin = 1.5A		915	1010	1110	A × Ω
Blocking MOSFET On-Resistance between VBUS and PMID (Q1)	R _{ON_QBLK}	T _J = +25°C			28	34.5	mΩ
		T _J = -40°C to +85°C			28	43.2	
D+/D- Detection							
D+/D- Voltage Source (600mV)	V _{D+D-_600MVSRC}			550	600	613	mV
D+ Current Source (10μA)	I _{D+_10UASRC}	V _{D+} = 200mV		7	10	16	μA
D+/D- Current Sink (100μA)	I _{D+D-_100UASNK}	V _{D+} = 500mV		74	100	150	μA
D+/D- Comparator Threshold for Secondary Detection	V _{D+D-_0P325}	D+ pin rising		278		324	mV
D- Resistor to Ground (19kΩ)	R _{D-_19K}	V _{D-} = 500mV		17.8		21.7	kΩ
D+ Comparator Threshold for Data Contact Detection	V _{D+_0P8}	D+ pin rising				855	mV
D+/D- Threshold for Non-Standard Adaptor	V _{D+D-_1P2}			1.04		1.43	V
	V _{D+D-_2P0}			1.88		2.1	
	V _{D+D-_2P8}			2.6		2.87	
D+/D- Leakage Current	I _{D+D-_LKG}	HIZ mode		-1		1	μA

I²C Controlled 2A, 2-Cell Boost Battery Charger SGM41529 with Power Path, OTG for USB Input

ELECTRICAL CHARACTERISTICS (continued)

($V_{VBUS_UVLOZ} < V_{VBUS} < V_{VBUS_OV}$, $T_J = -40^{\circ}\text{C}$ to $+85^{\circ}\text{C}$, typical values are at $T_J = +25^{\circ}\text{C}$, unless otherwise noted.)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
Battery Over-Voltage Protection						
Battery Over-Voltage Rising Threshold	V _{BAT_OVP_RISING}	V _{BAT} rising, as percentage of V _{REG}	102.3	104	105.6	%
Battery Over-Voltage Falling Threshold	V _{BAT_OVP_FALLING}	V _{BAT} falling, as percentage of V _{REG}	100.4	102	103.6	%
Thermal Regulation and Thermal Shutdown						
Junction Temperature Regulation Threshold	T _{REG}	T _{REG} [1:0] = 11 (+120°C)		120		°C
Thermal Shutdown Rising Threshold	T _{SHUT_RISING}	Temperature increasing		150		°C
Thermal Shutdown Falling Threshold		Temperature decreasing		120		°C
JEITA Thermistor Comparator (Boost Mode)						
T1 (0°C) Threshold Voltage on TS Pin	V _{T1}	As percentage to V _{REGN} , charge suspended below T1	73.01	73.25	73.6	%
V _{T1} Falling	V _{T1_HYS}	As percentage to V _{REGN} , charge re-enabled to I _{CHG} /5 (JEITA_ISETC[1:0] = 01) and V _{REG} above T1		1.3		%
T2 (10°C) Threshold Voltage on TS Pin	V _{T2}	As percentage to V _{REGN} , charge set to I _{CHG} /5 (JEITA_ISETC[1:0] = 01) and V _{REG} below T2	67.98	68.25	68.58	%
V _{T2} Falling	V _{T2_HYS}	As percentage to V _{REGN} , charge set to I _{CHG} and V _{REG} above T2		1.2		%
T3 (45°C) Threshold Voltage on TS Pin	V _{T3}	As percentage to V _{REGN} , charge set to I _{CHG} (JEITA_ISETH = 1) and 8V (JEITA_VSET[1:0] = 01) above T3	44.39	44.75	44.88	%
V _{T3} Rising	V _{T3_HYS}	As percentage to V _{REGN} , charge set to I _{CHG} and V _{REG} below T3		1		%
T4 (60°C) Threshold Voltage on TS Pin	V _{T4}	As percentage to V _{REGN} , charge suspended above T4	33.94	34.375	34.52	%
V _{T4} Rising	V _{T4_HYS}	As percentage to V _{REGN} , charge set to I _{CHG} (JEITA_ISETH = 1) and 8V (JEITA_VSET[1:0] = 01) below T4		1.35		%
Cold/Hot Thermistor Comparator (OTG Buck Mode)						
Cold Temperature Threshold 0 (TS Pin Voltage Rising Threshold)	V _{BCOLD0}	As percentage to V _{REGN} , BCOLD = 0 (Approx. -10°C w/ 103AT)	76.7	77	77.4	%
Cold Temperature Threshold 0 (TS Pin Voltage Falling Threshold)	V _{BCOLD0_HYS}	As percentage to V _{REGN}		1		%
Cold Temperature Threshold 1 (TS Pin Voltage Rising Threshold)	V _{BCOLD1}	As percentage to V _{REGN} , BCOLD = 1 (Approx. -20°C w/ 103AT)	79.7	80	80.4	%
Cold Temperature Threshold 1 (TS Pin Voltage Falling Threshold)	V _{BCOLD1_HYS}	As Percentage to V _{REGN}		1		%
Hot Temperature Threshold 0 (TS Pin Voltage Falling Threshold)	V _{BHOT0}	As percentage to V _{REGN} , BHOT[1:0] = 01 (Approx. 55°C w/ 103AT)	37.31	37.75	38.06	%
Hot Temperature Threshold 0 (TS Pin Voltage Rising Threshold)	V _{BHOT0_HYS}	As Percentage to V _{REGN}		3		%
Hot Temperature Threshold 1 (TS Pin Voltage Falling Threshold)	V _{BHOT1}	As percentage to V _{REGN} , BHOT[1:0] = 00 (Approx. 60°C w/ 103AT)	33.94	34.75	34.63	%
Hot Temperature Threshold 1 (TS Pin Voltage Rising Threshold)	V _{BHOT1_HYS}	As percentage to V _{REGN}		3		%
Hot Temperature Threshold 2 (TS Pin Voltage Falling Threshold)	V _{BHOT2}	As percentage to V _{REGN} , BHOT[1:0] = 10 (Approx. 65°C w/ 103AT)	30.74	31.25	31.48	%
Hot Temperature Threshold 2 (TS Pin Voltage Rising Threshold)	V _{BHOT2_HY2}	As percentage to V _{REGN}		3		%
Boost Mode Converter						
PWM Switching Frequency	f _{SW}	Oscillator frequency	1.35	1.5	1.65	MHz

SGM41529

I²C Controlled 2A, 2-Cell Boost Battery Charger with Power Path, OTG for USB Input

ELECTRICAL CHARACTERISTICS (continued)

($V_{VBUS_UVLOZ} < V_{VBUS} < V_{VBUS_OV}$, $T_J = -40^{\circ}\text{C}$ to $+85^{\circ}\text{C}$, typical values are at $T_J = +25^{\circ}\text{C}$, unless otherwise noted.)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
OTG Buck Mode Converter						
Battery Voltage Exiting OTG Mode	V_{OTG_BAT}	V_{BAT} falling	5.84	6	6.11	V
Typical OTG Buck Mode Voltage Regulation Range	V_{OTG_RANGE}		4.5		5.5	V
Typical OTG Buck Mode Voltage Regulation Step	V_{OTG_STEP}			100		mV
OTG Buck Mode Voltage Regulation Accuracy	V_{OTG_ACC}	$I_{VBUS} = 0\text{A}$, $OTG_VLIM[3:0] = 5.1\text{V}$	-3		3	%
Typical OTG Buck Mode Current Regulation Range	I_{OTG_RANGE}		0.5		2	A
Typical OTG Buck Mode Current Regulation Step	I_{OTG_STEP}			100		mA
OTG Buck Mode Current Regulation Accuracy	I_{OTG_ACC}	$OTG_ILIM[3:0] = 1\text{A}$		-5		%
OTG Buck Mode Over-Voltage Threshold	V_{OTG_OVP}		5.8	6		V
REGN LDO						
REGN LDO Output Voltage	V_{REGN}	$V_{VBUS} = 5\text{V}$, $I_{REGN} = 20\text{mA}$	4.54	4.7	4.79	V
REGN LDO Current Limit	I_{REGN}	$V_{VBUS} = 5\text{V}$, $V_{REGN} = 3.8\text{V}$	45			mA
Analog-to-Digital Converter (ADC)						
Conversion Time, Each Measurement	t_{ADC_CONV}	$ADC_SAMPLE[1:0] = 00$		8.8		ms
		$ADC_SAMPLE[1:0] = 01$		4.7		
		$ADC_SAMPLE[1:0] = 10$		2.7		
		$ADC_SAMPLE[1:0] = 11$		1.6		
Effective Resolution	ADCRRES	$ADC_SAMPLE[1:0] = 00$	14	15		bits
		$ADC_SAMPLE[1:0] = 01$	13	14		
		$ADC_SAMPLE[1:0] = 10$	12	13		
		$ADC_SAMPLE[1:0] = 11$	10	12		
ADC VBUS Current Range	$I_{BUS_ADC_RANGE}$		0		4	A
ADC VBUS Current LSB	$I_{BUS_ADC_LSB}$			1		mA
ADC BAT Current Range	$I_{BAT_ADC_RANGE}$		0		4	A
ADC BAT Current LSB	$I_{BAT_ADC_LSB}$			1		mA
ADC VBUS Voltage Range	$V_{VBUS_ADC_RANGE}$		0		6.5	V
ADC VBUS Voltage LSB	$V_{VBUS_ADC_LSB}$			1		mV
ADC SYS Voltage Range	$V_{SYS_ADC_RANGE}$		0		10	V
ADC SYS Voltage LSB	$V_{SYS_ADC_LSB}$			1		mV
ADC BAT Voltage Range	$V_{BAT_ADC_RANGE}$		0		10	V
ADC BAT Voltage LSB	$V_{BAT_ADC_LSB}$			1		mV
ADC TS Voltage Range	$V_{TS_ADC_RANGE}$		20		80	%
ADC TS Voltage LSB	$V_{TS_ADC_LSB}$			0.098		%
ADC Die Temperature Range	$V_{TDIE_ADC_RANGE}$		0		150	$^{\circ}\text{C}$
ADC Die Temperature LSB	$V_{TDIE_ADC_LSB}$			0.5		$^{\circ}\text{C}$
I²C Interface Characteristics (SCL, SDA)						
Input High Threshold (SDA and SCL)	V_{IH}	Pull up rail 1.8V	1.3			V
Input Low Threshold	V_{IL}	Pull up rail 1.8V			0.4	V
Output Low Threshold	V_{OL}	Sink current = 5mA			0.4	V
High-Level Leakage Current	I_{BIAS}	Pull up rail 1.8V			1	μA

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ELECTRICAL CHARACTERISTICS (continued)

($V_{VBUS_UVLOZ} < V_{VBUS} < V_{VBUS_OV}$, $T_J = -40^{\circ}\text{C}$ to $+85^{\circ}\text{C}$, typical values are at $T_J = +25^{\circ}\text{C}$, unless otherwise noted.)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
Logic I/O Pin Characteristics (nCE)						
Input High Threshold	V_{IH_CEZ}		1.3			V
Input Low Threshold	V_{IL_CEZ}				0.4	V
High-Level Leakage Current	$I_{IN_BIAS_CEZ}$	Pull up rail 1.8V			1	μA
Logic Output Pin Characteristics (nINT, nPG, STAT)						
Output Low Threshold	V_{OL}	Sink current = 5mA			0.4	V
High-Level Leakage Current	I_{OUT_BIAS}	Pull up rail 1.8V			1	μA

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TIMING REQUIREMENTS

($V_{VBUS_UVLOZ} < V_{VBUS} < V_{VBUS_OV}$, $T_J = -40^{\circ}\text{C}$ to $+85^{\circ}\text{C}$, typical values are at $T_J = +25^{\circ}\text{C}$, unless otherwise noted.)

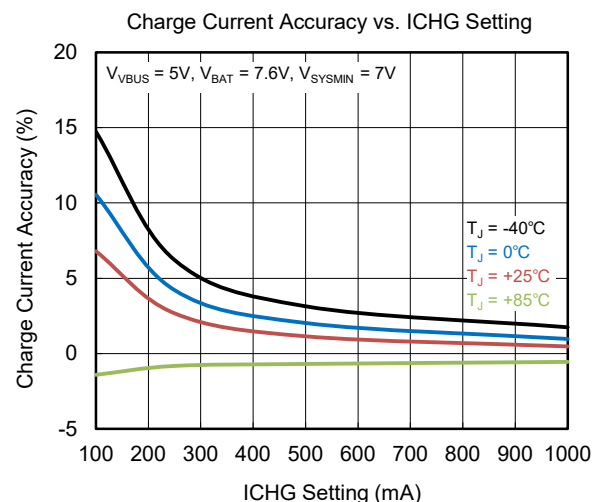
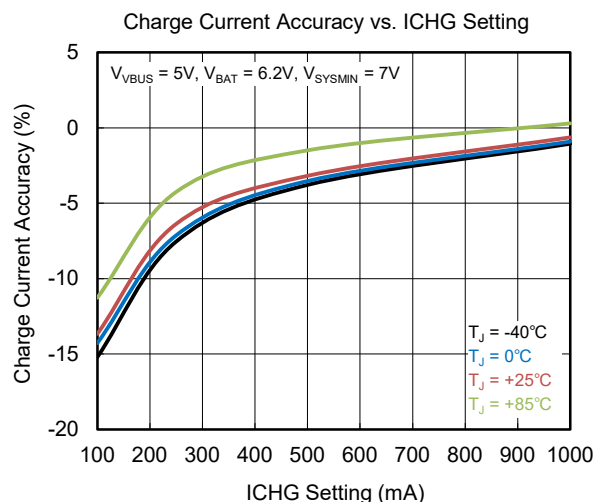
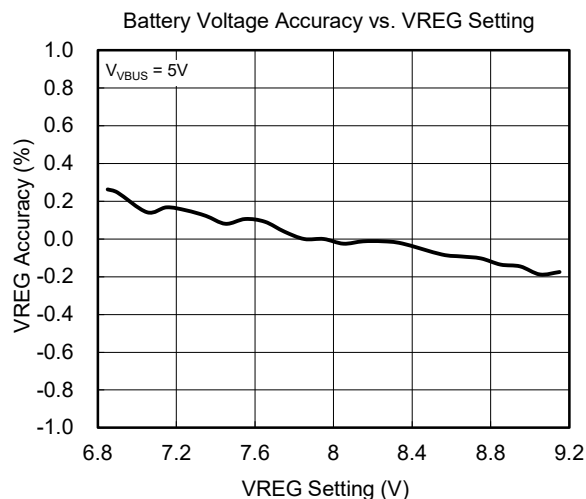
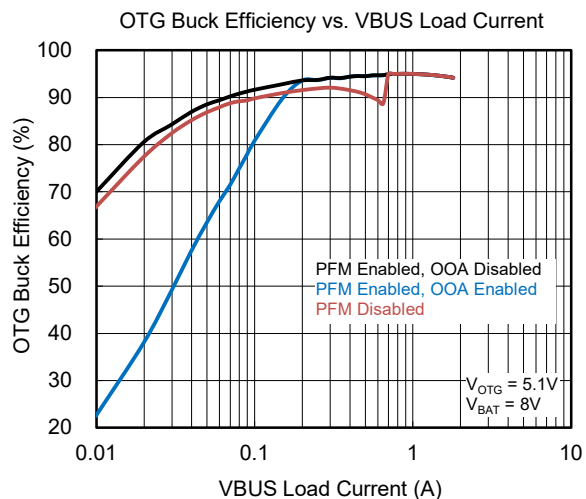
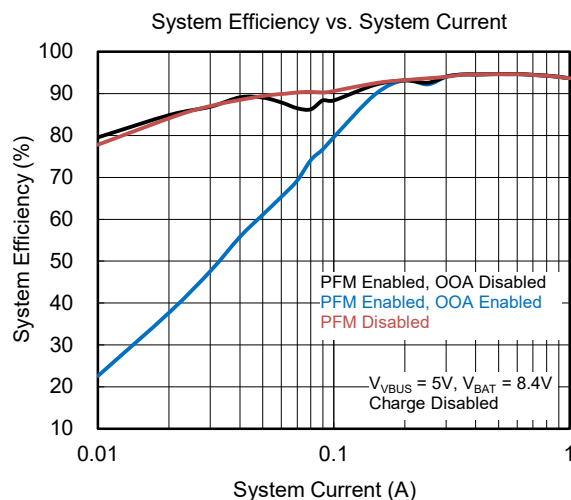
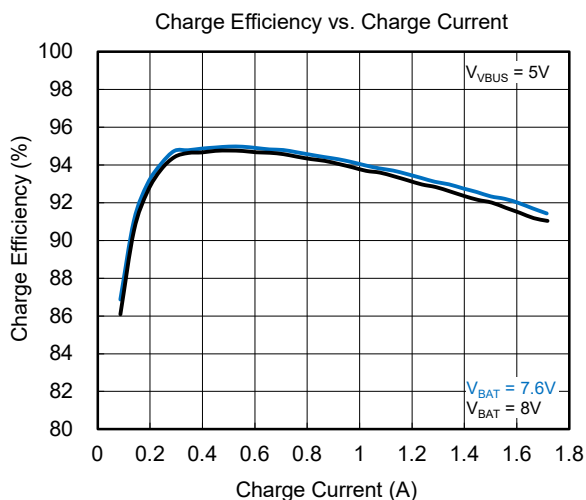
PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
V_{VBUS}/V_{BAT} Power-Up						
VBUS OVP Reaction Time	t_{VBUS_OV}	V _{VBUS} rising above V _{VBUS_OV} threshold to turn off the converter		200		ns
Wait Window for Bad Adaptor Detection	t_{BAD_SRC}			30		ms
Battery Charger						
Deglitch Time for Charge Termination	t_{TERM_DGL}	Charge current falling below I _{TERM}		250		ms
Deglitch Time for Recharge Threshold	t_{RECHG_DGL}	BAT voltage falling below V _{RECHG} = 100mV		250		ms
Deglitch Time for Battery Over-Voltage to Disable Charge	$t_{BAT_OVP_DGL}$			1		μs
Typical Charge Safety Timer Range	t_{SAFETY}	CHG_TIMER[1:0] = 10 (16.5h)	13.9	16.5	20	h
Typical Top-Off Timer Range	t_{TOP_OFF}	TOPOFF_TIMER[1:0] = 10 (30min)	29	36	43	min
I²C Interface						
SCL Clock Frequency	f_{SCL}				1000	kHz
Digital Clock and Watchdog Timer						
Watchdog Reset Time	t_{WDT}	WATCHDOG[1:0] = 11 (160s)	REGN LDO disabled		164	s
			REGN LDO enabled	133	164	
Digital Clock Frequency in Low Power	f_{LPDIG}	REGN LDO disabled		30		kHz
Digital Clock Frequency	f_{DIG}	REGN LDO enabled		1.5		MHz

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TYPICAL PERFORMANCE CHARACTERISTICS

$T_J = +25^\circ\text{C}$, $C_{VBUS} = 1\mu\text{F}$, $C_{PMID} = 10\mu\text{F}$, $C_{BAT} = 10\mu\text{F}$, $C_{SYS} = 2 \times 22\mu\text{F}$, $L = 1\mu\text{H}$, unless otherwise noted.

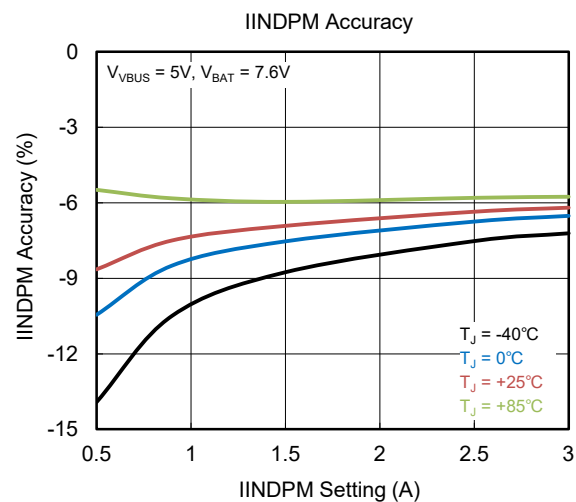
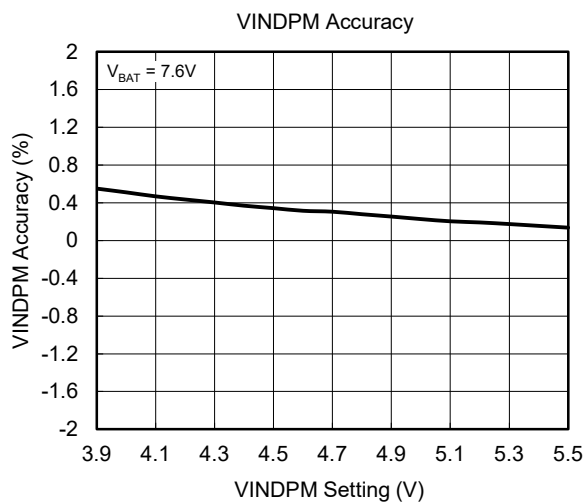
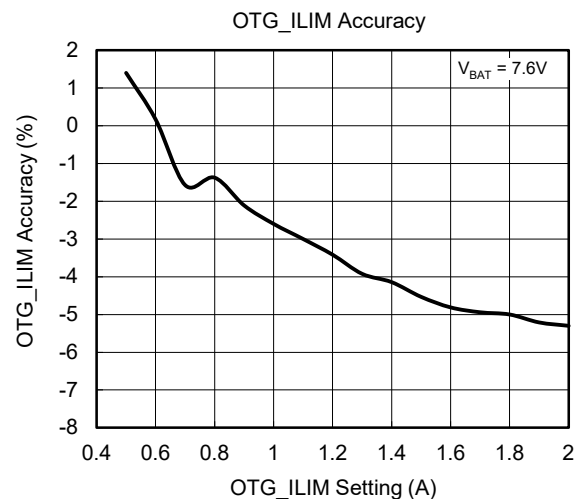
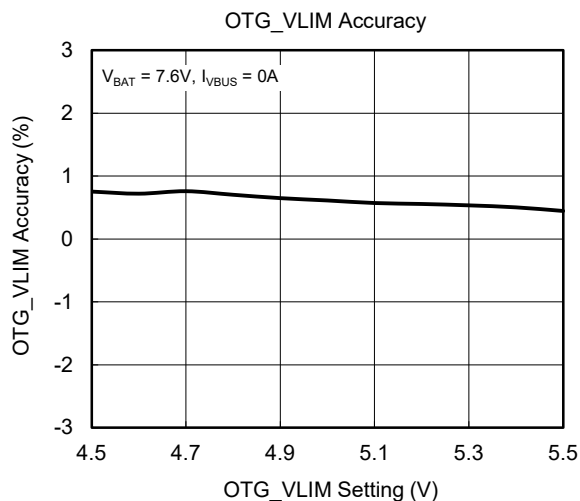
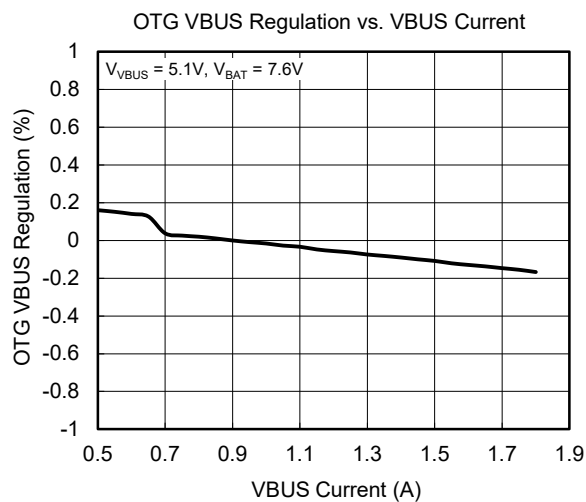
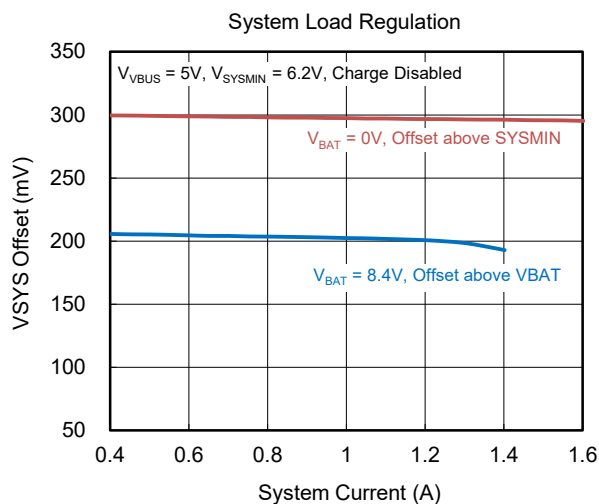


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I²C Controlled 2A, 2-Cell Boost Battery Charger with Power Path, OTG for USB Input

TYPICAL PERFORMANCE CHARACTERISTICS (continued)

$T_J = +25^\circ\text{C}$, $C_{VBUS} = 1\mu\text{F}$, $C_{PMID} = 10\mu\text{F}$, $C_{BAT} = 10\mu\text{F}$, $C_{SYS} = 2 \times 22\mu\text{F}$, $L = 1\mu\text{H}$, unless otherwise noted.

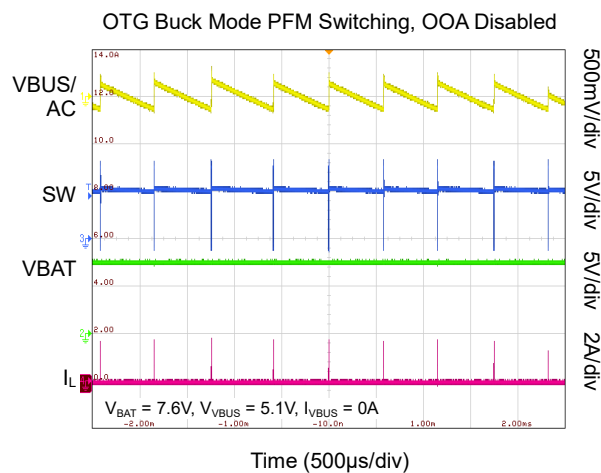
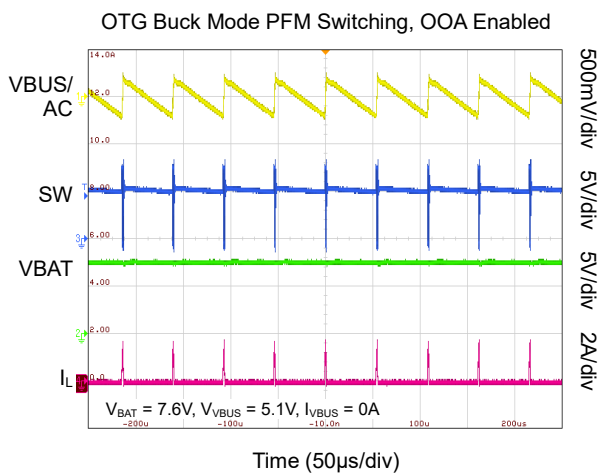
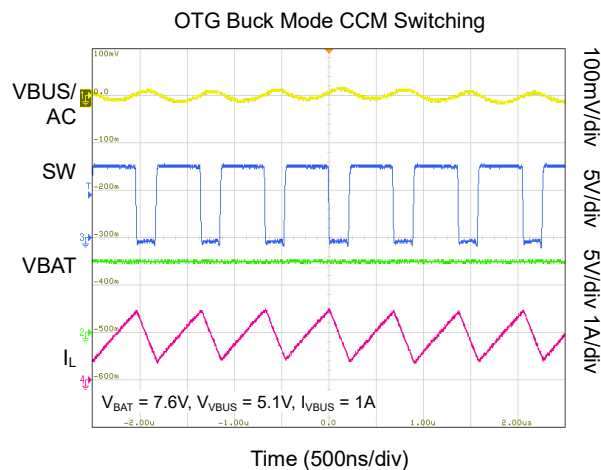
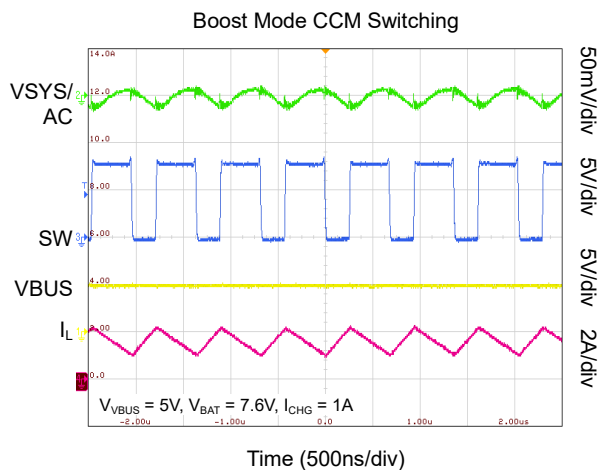
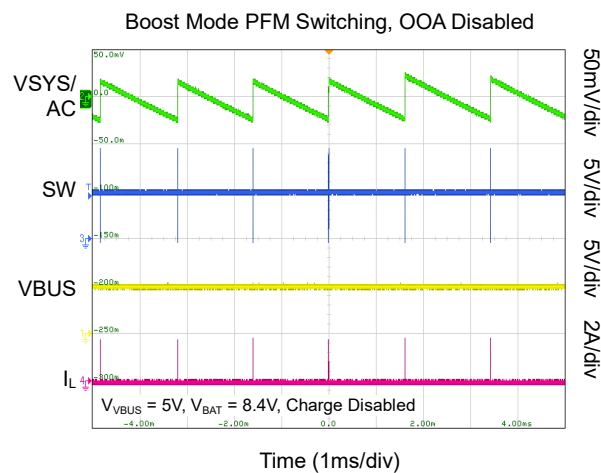
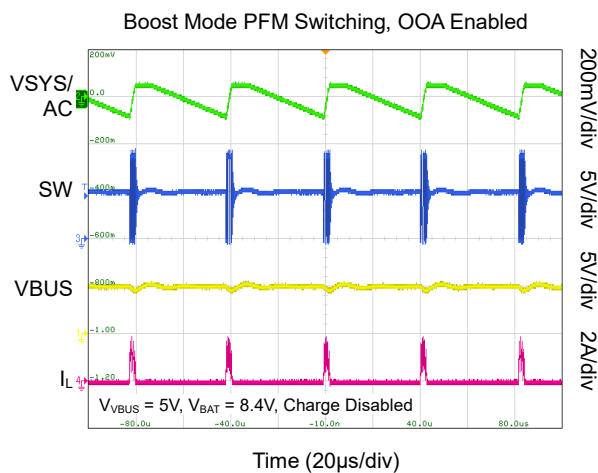


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TYPICAL PERFORMANCE CHARACTERISTICS (continued)

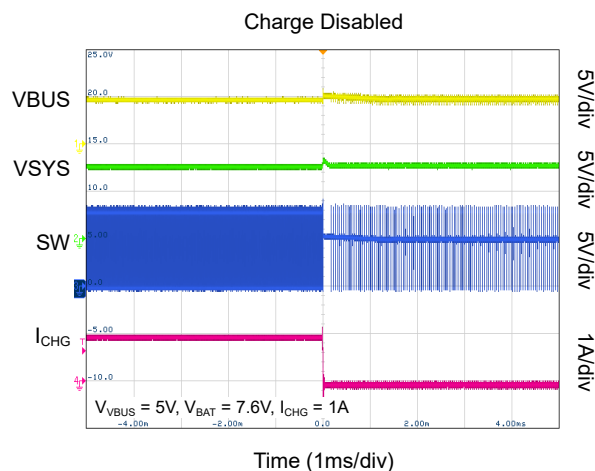
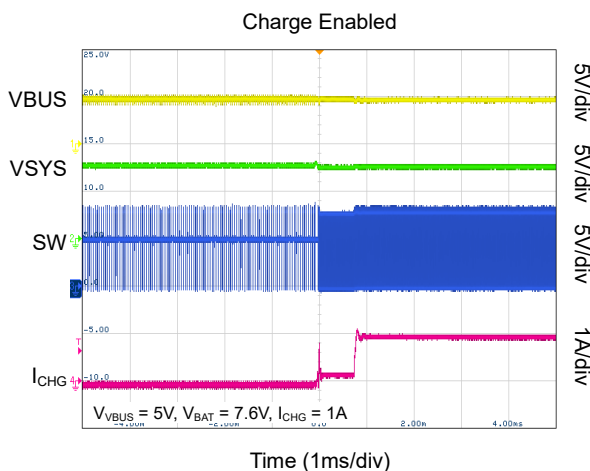
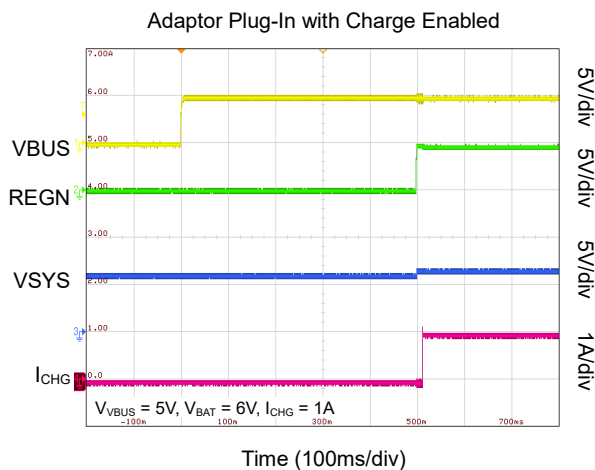
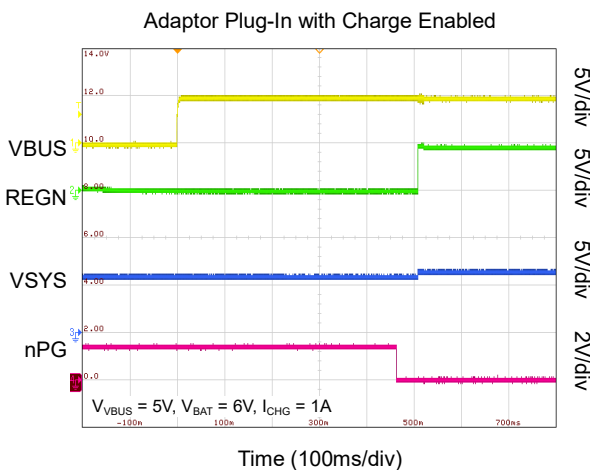
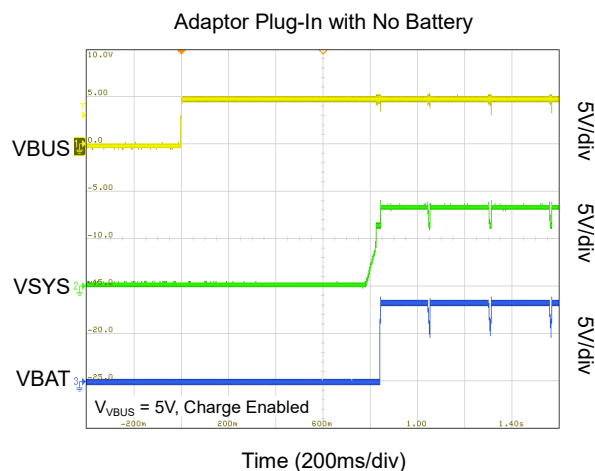
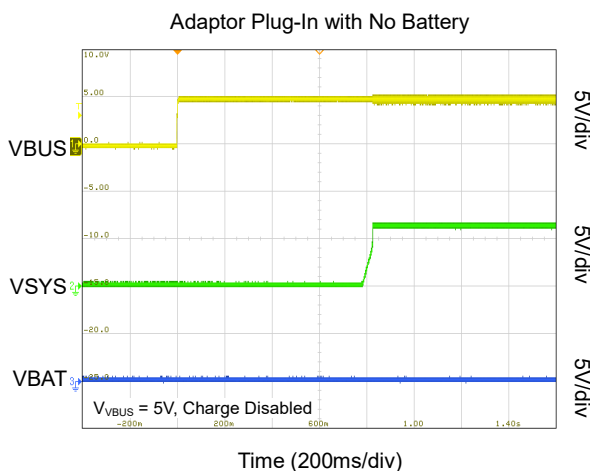
$T_J = +25^{\circ}\text{C}$, $C_{VBUS} = 1\mu\text{F}$, $C_{PMID} = 10\mu\text{F}$, $C_{BAT} = 10\mu\text{F}$, $C_{SYS} = 2 \times 22\mu\text{F}$, $L = 1\mu\text{H}$, unless otherwise noted.



SGM41529 I²C Controlled 2A, 2-Cell Boost Battery Charger with Power Path, OTG for USB Input

TYPICAL PERFORMANCE CHARACTERISTICS (continued)

$T_J = +25^\circ\text{C}$, $C_{VBUS} = 1\mu\text{F}$, $C_{PMID} = 10\mu\text{F}$, $C_{BAT} = 10\mu\text{F}$, $C_{SYS} = 2 \times 22\mu\text{F}$, $L = 1\mu\text{H}$, unless otherwise noted.

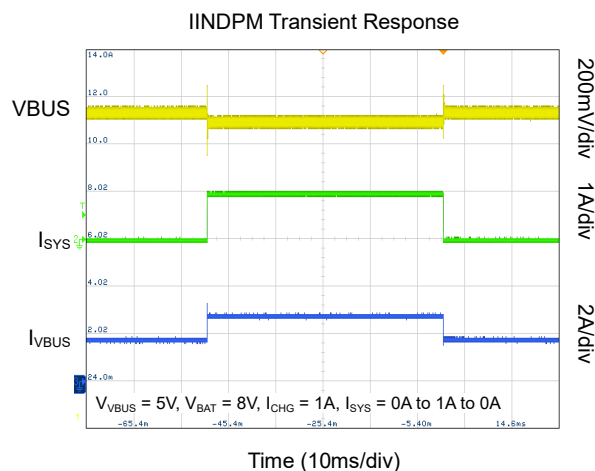
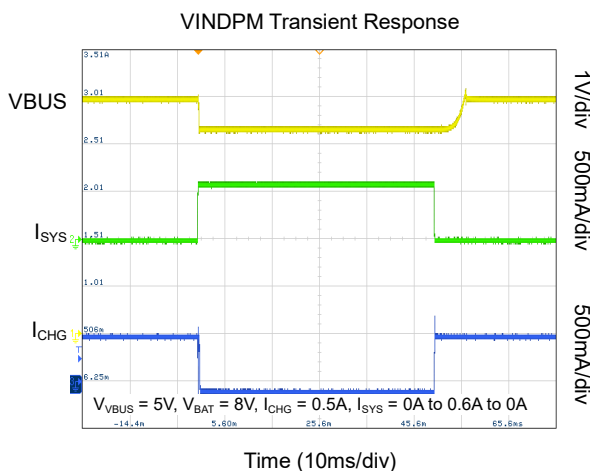
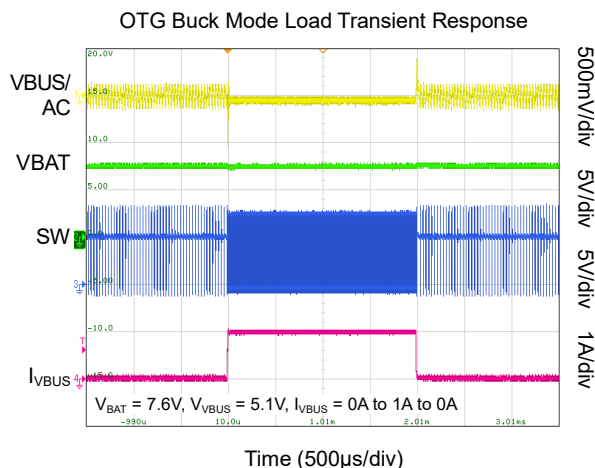
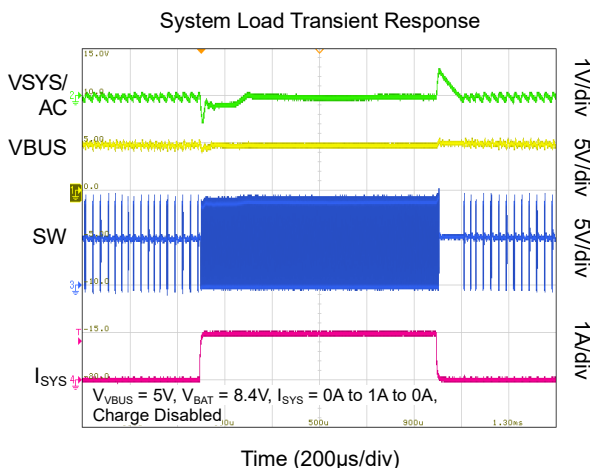
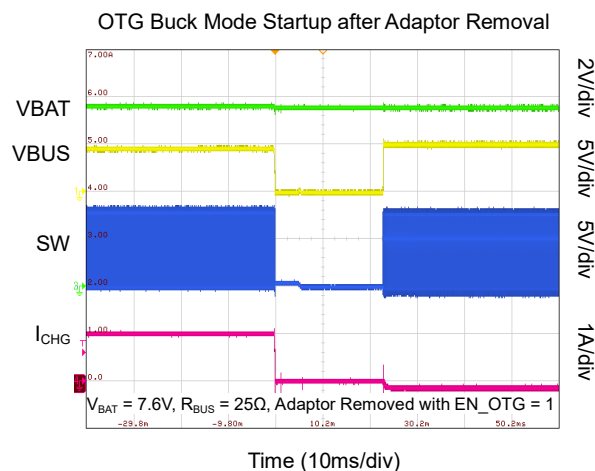
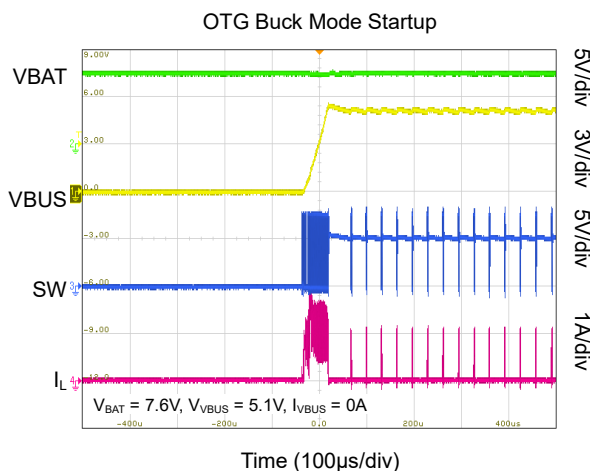


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I²C Controlled 2A, 2-Cell Boost Battery Charger with Power Path, OTG for USB Input

TYPICAL PERFORMANCE CHARACTERISTICS (continued)

$T_J = +25^{\circ}\text{C}$, $C_{VBUS} = 1\mu\text{F}$, $C_{PMID} = 10\mu\text{F}$, $C_{BAT} = 10\mu\text{F}$, $C_{SYS} = 2 \times 22\mu\text{F}$, $L = 1\mu\text{H}$, unless otherwise noted.



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I²C Controlled 2A, 2-Cell Boost Battery Charger with Power Path, OTG for USB Input

TYPICAL APPLICATION CIRCUIT

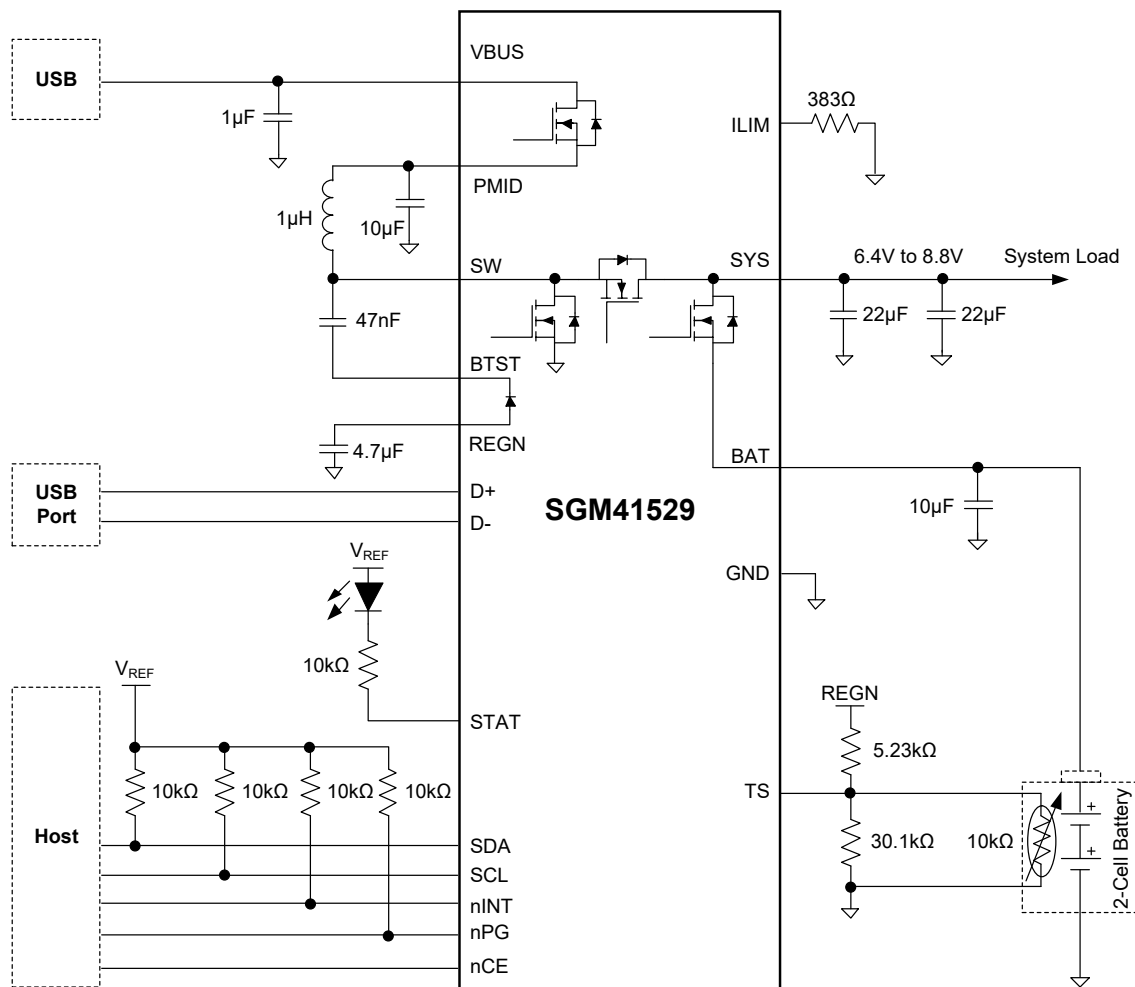


Figure 1. Typical Application Circuit

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I²C Controlled 2A, 2-Cell Boost Battery Charger with Power Path, OTG for USB Input

FUNCTIONAL BLOCK DIAGRAM

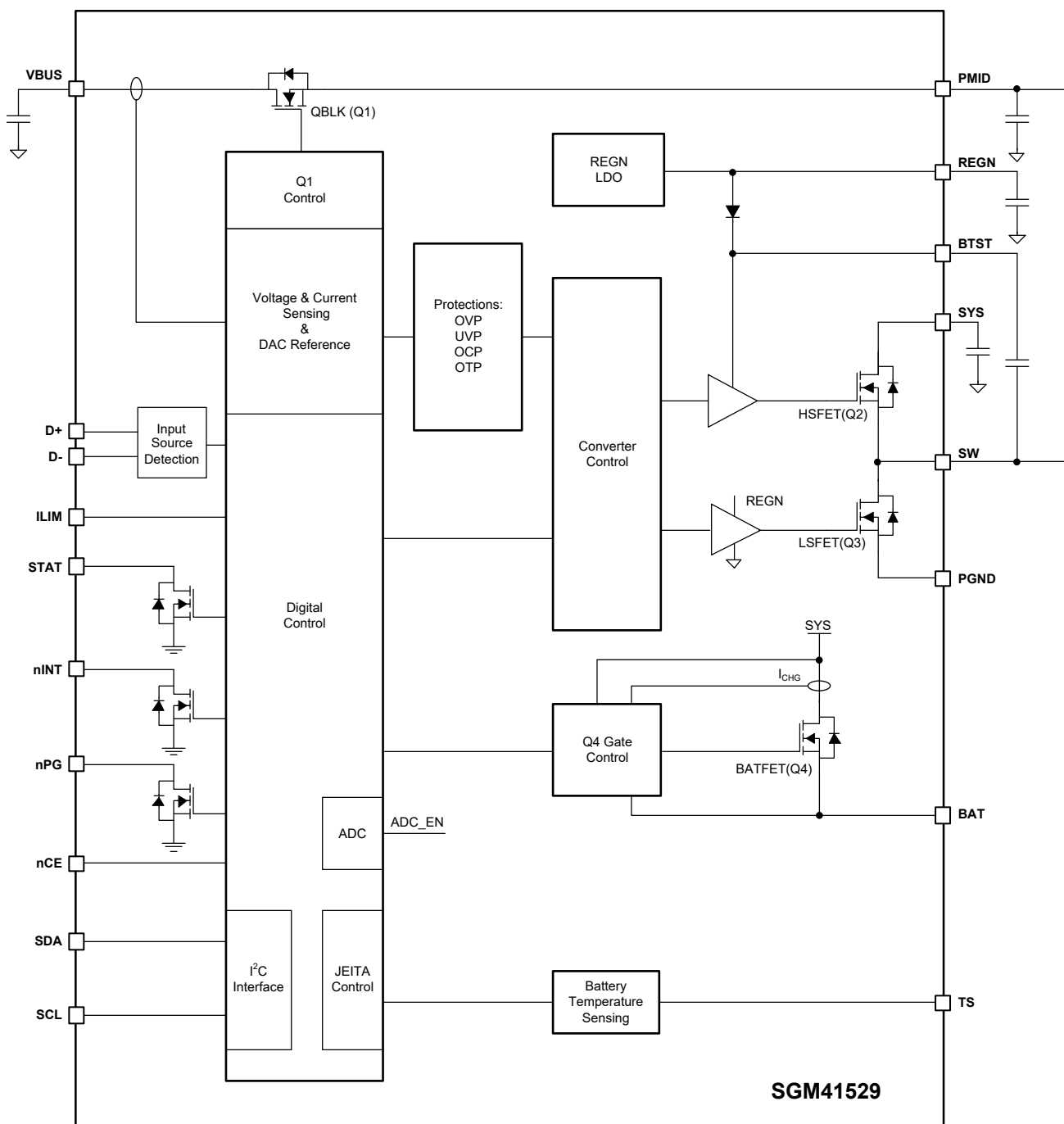


Figure 2. Block Diagram

SGM41529

I²C Controlled 2A, 2-Cell Boost Battery Charger with Power Path, OTG for USB Input

DETAILED DESCRIPTION

The SGM41529 is a battery charger and system power path management device with integrated Boost converter and power switches for using with 2-cell Li-Ion or Li-polymer batteries. The device includes four main power switches: input blocking MOSFET (Q1, QBLK), high-side switching MOSFET (Q2, HSFET), low-side switching MOSFET (Q3, LSFET), and battery MOSFET (Q4, BATFET).

Power-On Reset (POR)

The internal circuit of the device is powered by V_{VBUS} or V_{BAT} when V_{VBUS} or V_{BAT} goes above its UVLO threshold. When $V_{VBUS} > V_{VBUS_UVLOZ}$ or $V_{BAT} > V_{BAT_UVLOZ}$, a POR happens and activates the BATFET driver. Upon activation, the I²C interface will also be ready for communication, and all the registers reset to their default values.

Power-Up from Battery Only (No Input Source)

When only the battery is presented as a source and its voltage is above the UVLO threshold (V_{BAT_UVLOZ}), the BATFET turns on and connects the battery to the system. The quiescent current is minimum because the REGN LDO remains off. Conduction losses are also low due to small $R_{DS(on)}$ of BATFET. Low losses help to extend the battery run time. The discharge current through BATFET is continuously monitored.

Power-Up Process from the Input Power Source

Upon connection of an input source (VBUS), the input source from VBUS pin is checked to turn on the internal REGN LDO regulator and the bias circuits (whether the battery is present or not). Before the Boost converter starts, the device is detected and set the input current limit threshold. The sequences of actions when VBUS as input source is powered up are:

1. Poor power source detection (qualification).
2. Input power source type detection. (Based on D+/D- inputs.)
3. REGN LDO power-up.
4. DC/DC converter power-up.

Details of the power-up steps are explained in the following sections.

Poor Power Source Detection (Qualification)

When valid VBUS is plugged in, the input power source (adaptor or other source) is checked for its type and current capacity. To start the Boost converter, the input (VBUS) must meet the following conditions:

1. $V_{VBUS} < V_{VBUS_OV}$.
2. $V_{VBUS} > V_{BAD_SRC} + V_{BAD_SRC_HYS}$ (100mV TYP) during t_{BAD_SRC} test period (30ms TYP) in which the I_{BAD_SRC} (15mA TYP) current is pulled from VBUS.

Once the VBUS over-voltage protection is detected, the SGM41529 will automatically retry the detection when the over-voltage fault condition disappears. If above condition 2 is detected (bad source adaptor), the SGM41529 retries the detection and enters HIZ mode (EN_HIZ bit is set to 1) after 7 consecutive failures. In HIZ mode, the system is supplied by the battery only. Resume the converter operation requires re-plugging the adaptor and/or toggling the EN_HIZ bit. When the adaptor is plugged in, the EN_HIZ bit is automatically reset to 0.

Input Power Source Type Detection

The input power source detection will run through the D+/D- lines after the adaptor passes the poor source qualification when the EN_AUTO_INDET bit is set to 1. The SGM41529 follows the USB Battery Charging Specification 1.2 (BC1.2) to detect input adaptor power source type via D+/D- lines. All the types of SDP/CDP/DCP and non-standard adaptor can be automatically detected and indicated. When the input power source type detection is completed, the statuses of some related registers and pins are updated as detailed below:

1. Input current limit register (IINDPM[4:0]) is changed to set the right current limit.
2. Change the input voltage limit register (VINDPM[4:0]) to default setting if EN_VINDPM_RST = 1. Keep VINDPM[4:0] register value unchanged if EN_VINDPM_RST = 0.
3. Change the VBUS_STAT[2:0] bits according to the input source type detection result.
4. The PG_STAT bit is set to 1 and the nPG pin is pulled to logic low.
5. The nINT pin is pulsed to notify the host.

The input current or the input voltage is always limited by the IINDPM[4:0] or VINDPM[4:0] register and the limit can be updated by the host if needed. Regardless of the input current optimizer (ICO) setting, the charger input current is always limited by the lower value of the IINDPM[4:0] register or the current limit set by ILIM pin.

DETAILED DESCRIPTION (continued)

The input power source type detection is ignored if $EN_AUTO_INDET = 0$. In this case, the $IINDPM[4:0]$ register remains unchanged and the $VBUS_STAT[2:0]$ bits keep 000 (no input). If EN_VINDPM_RST bit is set to 0 by the host before the input power source type detection, the $VINDPM[4:0]$ register remains unchanged.

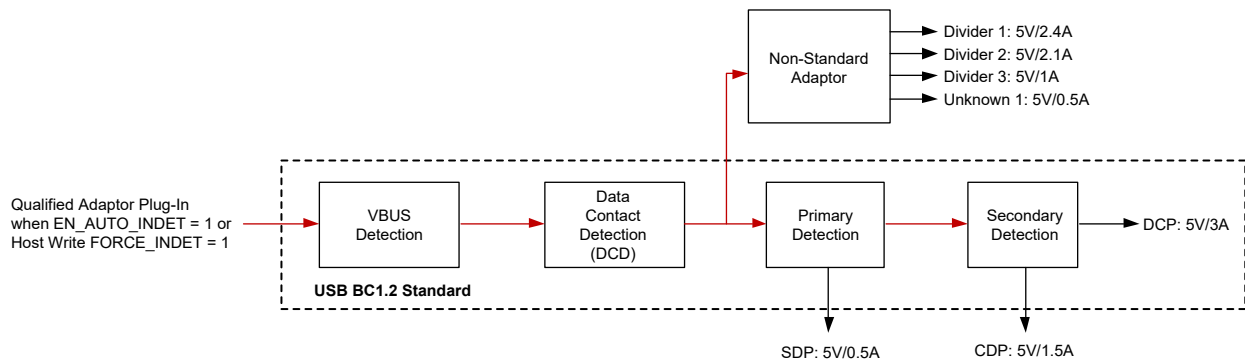
Input Current Limit by D+/D- Detection

The input current limit of SGM41529 is determined and set by the integrated D+/D- based input power source detection. Four major steps are included in the D+/D- detection: VBUS detection, data contact detection (DCD, detect non-standard adaptor), primary detection (detect SDP), and secondary detection (detect CDP and DCP). Please refer to Figure 3. Table 1 shows the non-standard adaptor type detection.

After the input power source type detection is completed, the $nINT$ pin sends out a low pulse to notify the host. In addition, the $VBUS_STAT[2:0]$ bits and the $IINDPM[4:0]$ register are updated as below, refer to Table 2.

Input Current Limit Force Detection

In host mode, the host can set $FORCE_INDET = 1$ to force the charger to run the input current limit detection. The $FORCE_INDET$ bit automatically reset to 0 once the detection is done. Due to the force detection result, the $VBUS_STAT[2:0]$ bits and $IINDPM[4:0]$ register may be changed.

**Figure 3. D+/D- Based USB BC1.2 Detection Flow****Table 1. Non-Standard Adaptor Type Detection**

Non-Standard Adaptor	D+ Threshold	D- Threshold	Input Current Limit (A)
Divider 1	V_{D+} within V_{2P8_VTH}	V_{D-} within V_{2P8_VTH}	2.4
Divider 2	V_{D+} within V_{2P8_VTH}	V_{D-} within V_{2P0_VTH}	2.1
Divider 3	V_{D+} within V_{2P0_VTH}	V_{D-} within V_{2P8_VTH}	1
Unknown 1	Unknown	Unknown	0.5

Table 2. Input Current Limit Setting and Status

D+/D- Detection	Input Current Limit (I_{INDPM})	$VBUS_STAT[2:0]$
USB SDP (USB500)	500mA	001
USB CDP	1.5A	010
USB DCP	3A	011
Divider 3	1A	110
Divider 2	2.1A	110
Divider 1	2.4A	110
Unknown 5V Adaptor	500mA	101

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DETAILED DESCRIPTION (continued)

REGN LDO Power-Up

The REGN low dropout regulator powers the internal bias circuits, HSFET and LSFET gate drivers and TS rail (thermistor pin). The nPG pin can also be pulled up to REGN. The REGN LDO enables when the following 3 conditions are satisfied and remain valid for 50ms delay time, otherwise the device stays in HIZ mode, and the REGN LDO keeps off.

1. $V_{VBUS} > V_{VBUS_UVLOZ}$ (in forward direction Boost mode).
2. A valid input source is detected by the poor power source detection.
3. Input power source type detection is done.

In HIZ state, the quiescent current drawn from VBUS is very small (I_{VBUS_HIZ}). System is powered only by the battery in HIZ mode.

DC/DC Converter Power-Up

The input current limit is set when input source detection is completed, then the VBUS_STAT[2:0] bits update to indicate the input source type, the nPG changes to logic low to indicate the power good and the PG_STAT and PG_FLAG bits also give the corresponding indication. The 1.5MHz switching converter composed of LSFET and HSFET is enabled and can start switching. Converter is initiated with a soft-start when the system voltage is ramped up.

The BATFET remains on to charge the battery if the battery charging function is enabled, otherwise BATFET turns off.

The SGM41529 provides an auto-run battery discharge source (I_{BAT_DISCHG} , 11.5mA (TYP)) for 13ms to detect the battery presence prior to the charge starting. During normal operation (include HIZ mode or battery only operation), this

discharge current can also be enabled by setting EN_BAT_DISCHG bit to 1.

At light load condition, if charging is disabled or the battery voltage is lower than V_{SYS_MIN} , the SGM41529 changes from PWM mode to PFM mode. The SYS and VBUS voltage relationship decides the switching duty cycle during the PFM operation. Host can disable the PFM operation via writing PFM_DIS = 1. If the PFM mode is selected, the out-of-audio (OOA) feature can be chosen under PFM_OOA_DIS = 0 condition. In OOA operation mode, the converter switching frequency is larger than 20kHz even at extremely light load to prevent the audible range operation.

Input Current Optimizer (ICO)

The SGM41529 provides input current optimizer (ICO) to identify the input adaptor source maximum power point. To avoid the input adaptor source overload and staying in VINDPM, the ICO algorithm identifies maximum input current limit of the adaptor automatically and updates this input current limit to ICO_ILIM[4:0] register.

The ICO function is default enabled, and it can be disabled by the host through setting EN_ICO bit to 0. After a DCP type power source is detected, the ICO algorithm runs automatically when EN_ICO bit is valid. The host can set FORCE_ICO bit to force the ICO algorithm under EN_ICO bit to 1 condition. Refer to Table 3.

The actual input current limit is reported by ICO_ILIM[4:0] register in ICO mode, while it is decided by the IINDPM[4:0] register when out of ICO mode. In addition, the actual input current is also limited by an external resistor at ILIM pin when EN_ILIM = 1.

Table 3. Automatic ICO Operation

Input Source	Input Current Limit (I_{INDPM})	Automatic ICO Mode when EN_ICO = 1
USB SDP (USB500)	500mA	Disable
USB CDP	1.5A	Disable
USB DCP	3A	Enable
Divider 3	1A	Disable
Divider 2	2.1A	Disable
Divider 1	2.4A	Disable
Unknown 5V Adaptor	500mA	Disable

DETAILED DESCRIPTION (continued)

When the ICO algorithm is activated, it runs to dynamically and continuously adjust the input current limit using ICO_ILIM[4:0] register. During the adjustment, the ICO_STAT[1:0] and ICO_FLAG bits change until they are finally set. The operation of ICO algorithm depends on the battery voltage as following:

Case 1: When $V_{BAT} < V_{SYS_MIN}$, the device starts ICO algorithm by ICO_ILIM[4:0] register with an initial value that equals the IINDPM. Where the IINDPM is the maximum input current limit that determined by the system.

Case 2: When $V_{BAT} > V_{SYS_MIN}$, the device starts ICO algorithm by ICO_ILIM[4:0] register with an initial value of 500mA. The 500mA is the minimum input current limit which minimizes the input power source overload.

During the optimization, if VINDPM is triggered, the ICO algorithm decreases the input current limit (the dynamic ICO_ILIM[4:0] register) to avoid input source overloading. When the maximum input current limit is detected, the ICO_ILIM[4:0] register reflects the optimal maximum input current limit which is not trigger VINDPM. The ICO_FLAG bit is set and the ICO_STAT[1:0] bits are updated to 10 to indicate the maximum input current detected.

In above case 1, if both VINDPM and IINDPM are not triggered at ICO_ILIM[4:0] initial value, the ICO_ILIM[4:0] register keeps the initial value and the ICO_STAT[1:0] = 01 to indicate the ICO optimization is in process. If the load becomes heavy, the VINDPM still not be triggered but IINDPM is triggered, the ICO algorithm is also completed, the ICO_ILIM[4:0] register keeps the initial value still, the ICO_FLAG bit is set and ICO_STAT[1:0] bits are updated to 10 to indicate the maximum input current detected.

In above case 2, if the VINDPM is not triggered and the converter is under light load condition, the ICO_ILIM[4:0] gives the input current limit a little higher than the actual input current (500mA minimum input current limit). The ICO_STAT[1:0] bits remain 01 to indicate the ICO optimization is in process. If the load becomes heavy, the ICO algorithm automatically runs to set new ICO_ILIM[4:0] register value.

Once the ICO algorithm is completed (ICO_STAT[1:0] = 10), the ICO_ILIM[4:0] register will keep un-change unless one of the following events occurs. Each of the following events can force the ICO algorithm to run again and reset the ICO_STAT[1:0] bits to 01:

1. Re-plugin the input source.
2. Host changes the IINDPM[4:0] register.
3. Host changes the VINDPM[4:0] register.
4. Host sets the FORCE_ICO bit to 1.
5. Resume from VBUS_OVP.

OTG Buck Mode

The SGM41529 supports USB On-The-Go (OTG). When a load device is connected to the USB port, the converter can operate as a Buck synchronous converter (Buck mode) with 1.5MHz switching frequency to supply power from the battery to that load. The USB OTG output current limit is programmable from 500mA to 2A (default). Converter will be set to Buck mode if at least 20ms is passed from enabling this mode (EN_OTG bit = 1) and the following conditions are satisfied:

1. $V_{BAT} > V_{OTG_BAT_RISE}$.
2. $V_{VBUS} < V_{VBUS_PRESENT}$.
3. TS pin voltage is out of V_{BHOTx} and V_{BCOLDx} range.

The default output Voltage is set to $V_{VBUS} = 5.1V$ (via OTG_VLIM[3:0] register) and is maintained as long as V_{BAT} is above V_{OTG_BAT} . The output current can reach up to the programmed value by OTG_ILIM[3:0] register (2A). The VBUS_STAT[2:0] status bits are set to 111 in Buck mode (OTG).

In OTG Buck mode, the SGM41529 default works in PFM mode at light load. The SYS and VBUS voltage relationship decides the switching duty cycle. Host can disable the OTG PFM operation via writing PFM_DIS bit. If the PFM mode is selected, the out-of-audio (OOA) feature can be chosen under PFM_OOA_DIS = 0 condition. In OOA operation mode, the converter switching frequency is larger than 20kHz even at extremely light load to prevent the audible range operation.

SGM41529 I²C Controlled 2A, 2-Cell Boost Battery Charger with Power Path, OTG for USB Input

DETAILED DESCRIPTION (continued)

Host Mode and Default Mode Operation with Watchdog Timer

After power-on reset, the SGM41529 starts in default mode (standalone) with all registers reset as default. If the watchdog timer is expired, the device will also enter default mode. When the host is in sleep mode or there is no host, the device stays in the default mode in which the SGM41529 operates like an autonomous charger. After 16.5 hours (default fast charging safety timer), the device stops charging the battery by turning off the BATFET, while the Boost converter continuously supplies the system load without shutdown.

Most of the flexibility features of the SGM41529 become available in the host mode when the device is controlled by a host with I²C. By setting the WD_RST bit to 1, the device changes to host mode. In this mode, the WD_STAT bit is low and all device parameters can be programmed. In order to prevent the device watchdog reset from going back to the default mode, the host must disable the watchdog timer by writing 00 to WATCHDOG[1:0], or consistently reset the watchdog timer before expiry by writing 1 to WD_RST bit to prevent WD_STAT bit from being set. Every time a 1 is written to WD_RST bit, the watchdog timer will restart counting. Therefore, it should be reset again before overflow (expiry) to keep the device in the host mode. The SGM41529 goes back to default mode and resets all related registers when the watchdog timer expires.

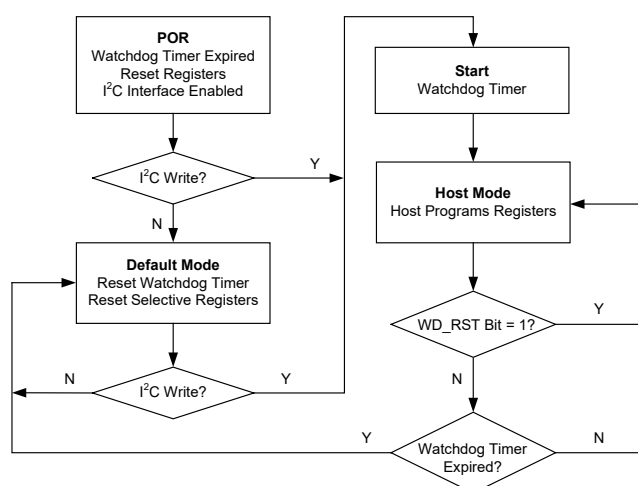


Figure 4. Watchdog Timer Flow Chart

Battery Charging Management

The SGM41529 is designed for charging 2-cell Li-Ion or Li-polymer batteries with a charge current up to 2.2A (MAX). The battery connection switch (BATFET) is in the charge or discharge current path and features low on-resistance to allow high efficiency and low voltage drop.

Charging Cycle in Autonomous Mode

Charging is enabled if EN_CHG = 1 and nCE pin is pulled low. In default mode, the SGM41529 runs a charge cycle with the default parameters itemized in Table 4. At any moment, the host can be controlled by changing to the host mode.

Table 4. Charging Parameter Default Settings

Default Mode	SGM41529
Charging Voltage (V _{REG})	8.4V
Fast Charging Current (I _{CHG})	1.5A
Pre-Charge Current (I _{PRECHG})	150mA
Termination Current (I _{TERM})	150mA
Temperature Profile	JEITA Enable
Charging Safety Timer	16.5h
Top-Off Timer	Disabled

Start a New Charging Cycle

If the converter can start switching and all the following conditions are satisfied, a new charge cycle starts:

- NTC temperature fault is not asserted (TS pin).
- Safety timer fault is not asserted.
- Charging enabled (2 conditions: EN_CHG bit = 1, nCE pin is low).
- Battery voltage is below the programmed full charge level (V_{REG}).

A new charge cycle starts automatically if battery voltage falls below the recharge threshold level (configured by VRECHG[1:0] bits). Also, if the charge cycle is finished, a new charging cycle can be initiated by toggling of the nCE pin or EN_CHG bit.

Normally, a charge cycle terminates when the charge voltage is above the recharge threshold level and the charging current falls below the termination threshold if the device is not in thermal regulation or dynamic power management (DPM) mode.

SGM41529

I²C Controlled 2A, 2-Cell Boost Battery Charger with Power Path, OTG for USB Input

DETAILED DESCRIPTION (continued)

Charge Status Report

The charge status bits (CHRG_STAT[2:0]) indicate the charging phases of the device as below:

- 000 = Not Charging
- 001 = Trickle Charge
- 010 = Pre-Charge
- 011 = Fast Charge (CC Mode)
- 100 = Taper Charge (CV Mode)
- 101 = Top-Off Timer Active
- 110 = Charge Termination

When the device changes to any of the above statuses, or the charge cycle is completed, the nINT pin is pulsed to notify the host.

In addition, the STAT pin output also indicates the charging status: LOW = charging, HIGH = charging complete or charge disabled, Blinking = charging fault.

Battery Charging Profile

The SGM41529 features a full battery charging profile with five phases. When a charging cycle starts, the battery voltage (V_{BAT}) is tested, and appropriate current and voltage regulation levels are selected as shown in Table 5. Depending on the detected status of the battery, the proper phase is selected to start or for continuation of the charging cycle. The five phases are trickle charge (battery voltage too low), pre-charge, constant current, constant voltage and an optional top-off trickle charging phase.

Note that in the DPM or thermal regulation modes, normal charging functions are temporarily modified: the charge current is less than the value in the register. The termination is disabled, and the charging safety timer is slowed down by counting at half clock rate.

Table 5. Charging Current Setting Based on V_{BAT}

V_{BAT} Voltage	Selected Charging Current	Default Value in the Register	CHRG_STAT[2:0]
$V_{BAT} < V_{BAT_SHORT}$	I_{BAT_SHORT}	80mA	001
$V_{BAT_SHORT} < V_{BAT} < V_{BAT_LOW}$	I_{PRECHG}	150mA	010
$V_{BAT_LOW} < V_{BAT} < V_{REG}$	I_{CHG}	1500mA	011

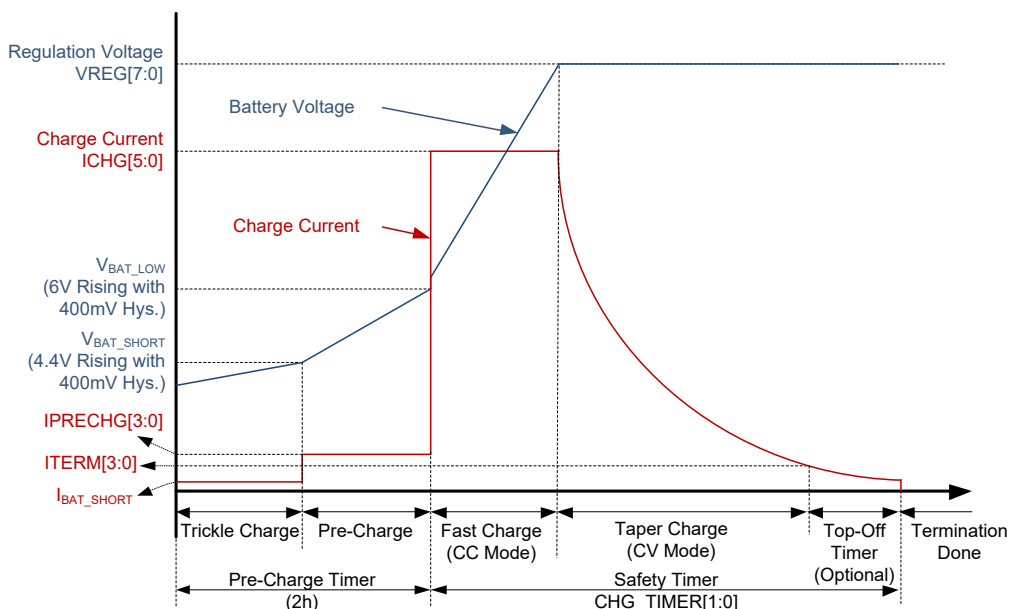


Figure 5. Battery Charging Profile

SGM41529

I²C Controlled 2A, 2-Cell Boost Battery Charger with Power Path, OTG for USB Input

DETAILED DESCRIPTION (continued)

Charge Termination

A charge cycle is terminated when the SGM41529 operates in the battery constant voltage regulation loop and the charge current falls below the programmed termination current. Unless there is a high power demand for system and it needs to operate in supplement mode, the BATFET turns off at the end of the charge cycle. Even after termination, the Boost converter operates continuously to supply the system.

CHRG_STAT[2:0] bits are set to 110, a negative pulse is sent to nINT pin and the STAT pin goes HIGH after termination.

If the charger is regulating input current, input voltage or junction temperature instead of charge current, termination will be temporarily prevented. EN_TERM bit is a termination control bit and can be set to 0 to permanently disable termination before it happens.

Due to offset of the internal current comparator, the termination charge current may be much higher (40% TYP) than the set value when it is set too low (50mA TYP). A delay in termination can be added (optional) as a compensation for comparator offset using a programmable top-off timer. During the delay, constant voltage charge phase continues and gives the falling charge current the chance to drop closer to the programmed value. The top-off delay timer has the same restrictions of the safety timer. In other words, if the safety timer is suspended under certain conditions, the top-off timer will also be suspended. And if the safety timer is slowed down, the top-off timer will also be slowed down. Code 101 in CHRG_STAT[2:0] indicates that the top-off timer is valid. The CHRG_STAT[2:0] bits change to 110 after the top-off timer expires. And the nINT pin reports a negative pulse to notify the host.

Any of the following events resets the top-off timer:

1. Disable to enable transition of nCE (charge enable).
2. A low to high change in the status of termination.
3. Set REG_RST bit to 1.

The setting of the top-off timer is applied at the time of termination detection and unless a new charge cycle is started, modifying the top-off timer parameters after termination has no effect. A negative pulse is sent to nINT when top-off timer is started or ended. If set CHRG_MASK bit to 1, the CHRG_STAT[2:0] bits change will not produce nINT pulse.

Temperature Qualification

The charging current and voltage of the battery must be limited when battery is cold or hot. A thermistor input for battery temperature monitoring is included in the device that can protect the battery based on JEITA guidelines. There is no battery temperature protection when Boost or Buck converter stops switching.

Compliance with JEITA Guideline

JEITA guideline (April 20, 2007 release) is implemented in the device for safe charging of the Li-Ion battery. JEITA highlights the considerations and limits that should be considered for charging at cold or hot battery temperatures. High charge current and voltage must be avoided outside, the normal operating temperatures (typically 0 °C and 60 °C). Four temperature levels are defined by JEITA from T1 (minimum) to T4 (maximum). Outside this range, charging should be stopped. The corresponding voltages sensed by NTC are named V_{T1} to V_{T4}. Due to the sensor negative resistance, a higher temperature results in a lower voltage on TS pin. The battery cool range is between T1 and T2, and the warm range is between T3 and T4. Charge must be limited in the cool and warm ranges.

One of the conditions for starting a charge cycle is having the TS voltage within V_{T1} to V_{T4} window limits. If the battery is too cold or too hot during charging and TS voltage exceeds the T1 - T4 limits, charging is suspended (zero charge current) and the controller waits for the battery temperature to come back within the T1 to T4 window.

JEITA recommends reducing charge current to 1/2 of fast charging current or lower at cool temperatures (T1 - T2). For warmer temperature (within T3 - T4 range), charge voltage is recommended to be kept below 4.1V/cell.

The SGM41529 follows the JEITA requirement by its flexible charge parameter settings. At warm temperature range (T3 - T4), the charge voltage can be set to 8.0V (default), 8.3V, V_{REG}, or charge suspend through JEITA_VSET[1:0] register. At cool temperatures (T1 - T2), the fast charge current can be set to 100%, 40%, or 20% (default) of I_{CHG} or charge suspend through JEITA_ISETC[1:0] bits. The charge termination is still enabled (if EN_TERM = 1) when the "cool" or "warm" temperature is detected.

DETAILED DESCRIPTION (continued)

A 103AT-2 type thermistor is recommended to use for the SGM41529. Other thermistors may be used and bias network (see Figure 6) can be calculated based on the following equations:

$$R_{T2} = \frac{V_{\text{REGN}} \times R_{\text{THCOLD}} \times R_{\text{THHOT}} \times \left(\frac{1}{V_{T1}} - \frac{1}{V_{T4}} \right)}{R_{\text{THHOT}} \times \left(\frac{V_{\text{REGN}}}{V_{T4}} - 1 \right) - R_{\text{THCOLD}} \times \left(\frac{V_{\text{REGN}}}{V_{T1}} - 1 \right)} \quad (1)$$

$$R_{T1} = \frac{\left(\left(\frac{V_{\text{REGN}}}{V_{T1}} \right) - 1 \right)}{\left(\frac{1}{R_{T2}} \right) + \left(\frac{1}{R_{\text{THCOLD}}} \right)} \quad (2)$$

Where, V_{T1} , V_{T4} and V_{REGN} are characteristics of the device, and R_{THCOLD} and R_{THHOT} are thermistor resistances (R_{TH}) at desired T_1 (Cold) and T_4 (Hot) temperatures. Select $T_{\text{COLD}} = 0^\circ\text{C}$ and $T_{\text{HOT}} = 60^\circ\text{C}$ for Li-Ion or Li-polymer batteries. For a 103AT-2 type thermistor $R_{\text{THCOLD}} = 27.28\text{k}\Omega$ and $R_{\text{THHOT}} = 3.02\text{k}\Omega$, the calculation results are: $R_{T1} = 5.24\text{k}\Omega$ and $R_{T2} = 30.31\text{k}\Omega$. The standard value of R_{T1} is $5.23\text{k}\Omega$ and that of R_{T2} is $30.1\text{k}\Omega$.

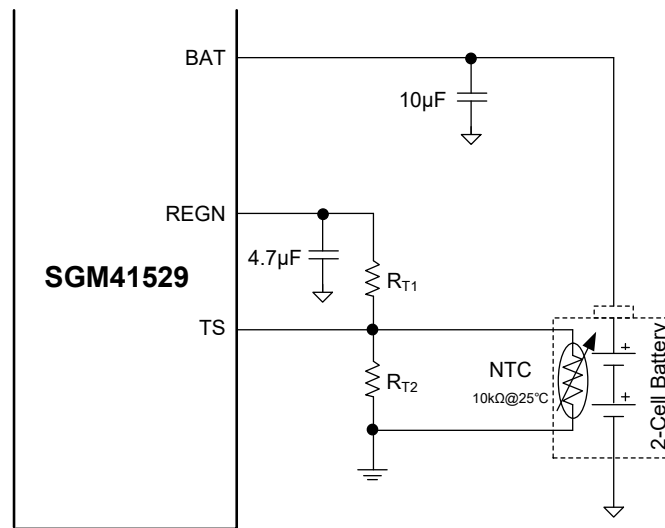


Figure 6. Battery Thermistor Connection and Bias Network

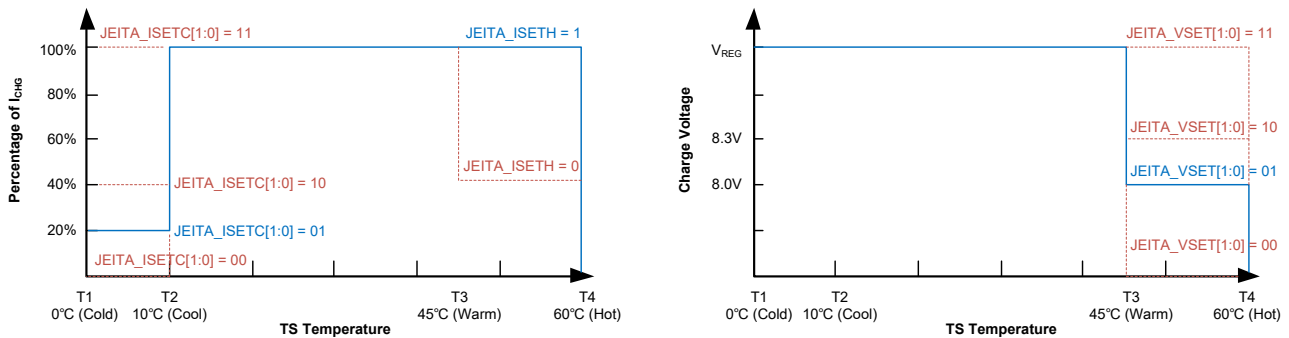


Figure 7. TS Based Current and Voltage Settings

DETAILED DESCRIPTION (continued)

OTG Buck Mode Temperature Monitoring

The device is capable of monitoring the battery temperature for safety during the OTG Buck mode. The temperature must remain within the V_{BCOLDx} to V_{BHOTx} thresholds, otherwise the OTG mode will be suspended and $VBUS_STAT[2:0]$ bits are set to 000. Moreover, $TS_STAT[2:0]$ bits are updated to report OTG mode cold or hot condition. Once the temperature returns within the right window, the OTG Buck mode is resumed and $TS_STAT[2:0]$ bits are cleared to 000 (normal).

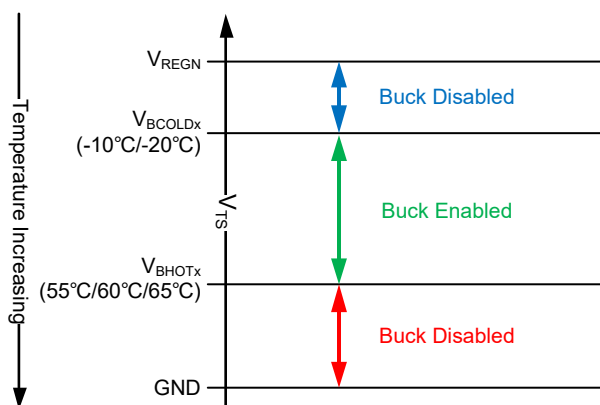


Figure 8. TS Pin Thermistor Temperature Window Settings in OTG Buck Mode

Safety Timer

Abnormal battery conditions may result in prolonged charge cycles. An internal safety timer is considered to stop charging in such conditions. If the charging safety timer timeout occurs, TMR_STAT bit is set and a negative pulse is sent to $nINT$ pin. This feature is optional and can be disabled by setting EN_TIMER bit to 0.

The safety timer counts at half clock rate when charger is under input voltage regulation, input current regulation or thermal regulation, and the actual charge current is always decreased. As an example, if the charging safety timer is set to 12 hours and the charger is under input current regulation ($IINDPM_STAT = 1$) in the whole charge cycle, the actual safety time will be 24 hours. Clearing the EN_TMR2X bit will disable the half clock rate feature. If $EN_TMR2X = 1$ and SGM41529 is already in DPM or thermal regulation, writing the $EN_TMR2X = 0$ will not take effect.

The safety timer is paused if supplement mode occurs or a fault occurs which disables the charging. The EN_TMR2X bit also has no effect in this condition because the timer counting

has stopped. It will resume once the fault condition is removed. If charging cycle is stopped and restarted by toggling nCE pin or EN_CHG bit, the timer resets and restarts a new timing.

The fast charge safety timer can be reset by the following events:

1. Stop and restart the charging cycle (change EN_CHG bit, toggle nCE pin, or charged battery falls below recharge threshold).
2. Charge status changes between pre-charge and fast charge (in default mode or host mode).

The pre-charge safety timer, including both trickle charge phase and pre-charge phase, is a fixed 2-hour counter and runs when $V_{BAT} < V_{BAT_LOW}$. It follows the same rules as the fast charge safety timer in terms of reset, getting suspended, and half-rate counting when EN_TMR2X is valid.

Narrow Voltage DC (NVDC) Design

The SGM41529 features an NVDC design using the BATFET that connects the system to the battery. By using the linear region of the BATFET, the charger regulates the system bus voltage (SYS pin) above the minimum setting using Boost converter even if the battery voltage is very low. MOSFET linear mode allows the large voltage difference between SYS and BAT pins to appear as V_{DS} across the switch while conducting and charging battery. $SYS_MIN[3:0]$ register sets the minimum system voltage (default 6.2V). If the system is in minimum system voltage regulation, $VSYS_STAT$ bit is set.

The BATFET operates in linear region when the battery voltage is lower than the minimum system voltage. The system voltage is regulated at 300mV (TYP) higher than the minimum system voltage. As the battery gradually gets charged, until its voltage rises above the minimum system voltage, the BATFET changes from linear mode to fully turned-on mode, and the system voltage keeps V_{DS} of BATFET higher than the battery voltage.

The system voltage is always regulated to 200mV (TYP) above the battery voltage if:

1. The charging is terminated.
2. Charging is disabled and the battery voltage is above the minimum system voltage setting.

DETAILED DESCRIPTION (continued)

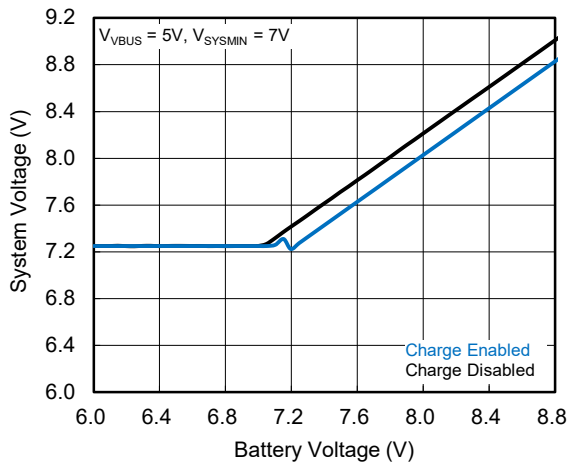


Figure 9. System Voltage vs. Battery Voltage

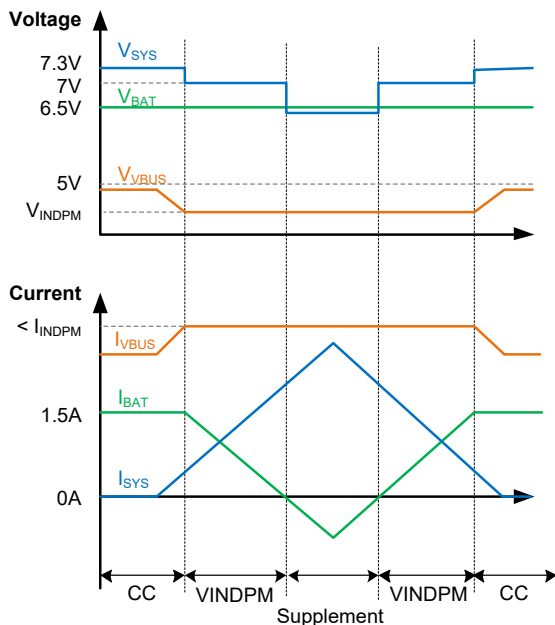
Dynamic Power Management (DPM)

The SGM41529 has a dynamic power management (DPM) feature. To implement DPM, the device always monitors the input current and voltage to regulate power demand from the source and avoid input adaptor overloading or to meet the maximum current limits specified in the USB specifications. Overloading an input power source may result in either the voltage tending to fall below the input voltage limit (V_{INDPM}) or

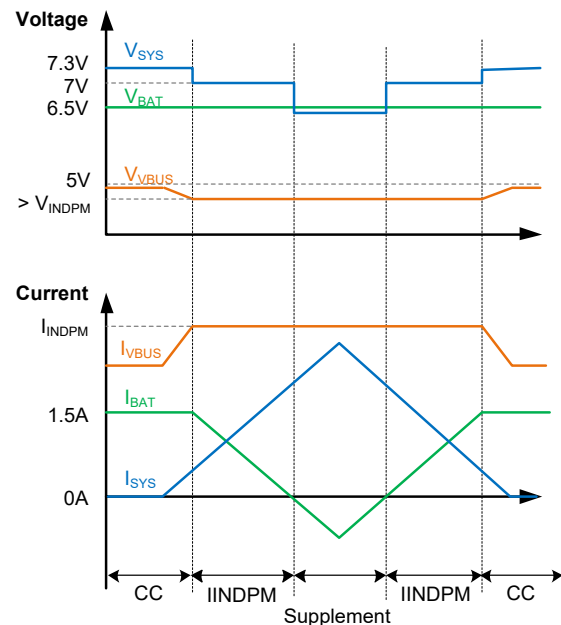
the current trying to exceed the input current limit (I_{INDPM} or $ICO_ILIM[4:0]$ or $ILIM$ pin setting). With DPM, the device keeps the V_{SYS} regulating to its minimum setting by reducing the battery charge current adequately such that the input parameter (voltage or current) does not exceed the limit. In other words, charge current is reduced to satisfy $I_{IN} \leq I_{INDPM}$ or $V_{IN} \geq V_{INDPM}$ whichever occurs first. DPM can be either an I_{IN} type (IINDPM) or V_{IN} type (VINDPM), depending on which limit is reached.

Changing to the supplement mode may be required if the charge current is decreased and reached to zero while the input is still overloaded. In this case, the charger reduces the system voltage below the battery voltage to allow operation in the supplement mode and provide a portion of system power demand from the battery through the BATFET.

The IINDPM_STAT or VINDPM_STAT status bits are set during an IINDPM or VINDPM respectively. Figure 10 summarizes the DPM behavior (IINDPM type) for a design example with a 5V/3A adaptor, 6.5V battery, 1.5A charge current setting and 7V minimum system voltage setting.



a) Input, Battery and System Voltage and Currents in VINDPM



b) Input, Battery and System Voltage and Currents in IINDPM

Figure 10. Input, Battery and System Voltage and Currents in DPM

DETAILED DESCRIPTION (continued)

Battery Supplement Mode

When the system is under heavy load, the system voltage may drop below the battery voltage, and the BATFET gradually starts to turn on. At low discharge currents, the BATFET gate voltage is regulated (R_{DS} modulation). At higher currents, the BATFET will turn fully on (reaching its lowest $R_{DS(on)}$). From this point, increasing the discharge current will linearly increase the BATFET V_{DS} (determined by $R_{DS(on)} \times I_D$). Using the MOSFET linear mode at lower currents prevents swinging oscillation from entering and exiting the supplement mode.

BATFET gate regulation V-I characteristic is shown in Figure 11. If the battery voltage falls below its minimum depletion, the BATFET turns off and exits supplement mode.

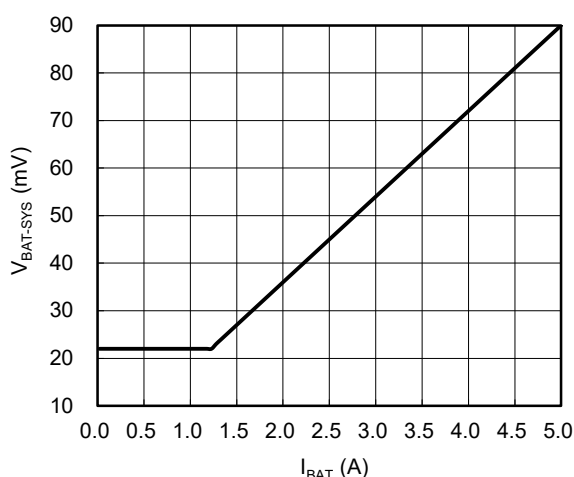


Figure 11. BATFET Gate Regulation V-I Curve

16-Bit ADC

The integrated 16-bit ADC in SGM41529 allows the user to get critical system information for optimizing the charger behavior. The ADC related functions are controlled by ADC control register. The EN_ADC bit gives the option to enable or disable the ADC for power save purpose. The ADC conversion behavior can be set to continuous or one-shot mode through the ADC_RATE bit. The EN_ADC bit is automatically cleared once a one-shot conversion cycle completes. Re-assert EN_ADC bit can start a new conversion.

Set EN_ADC bit to 1 to enable the ADC. The ADC can be operated when VBAT or VBUS is valid ($V_{BAT} > V_{BAT_UVLOZ}$ or $V_{VBUS_UVLOZ} < V_{VBUS} < V_{VBUS_OV}$). The SGM41529 will not execute the ADC under both VBAT and VBUS are invalid, and the ADC result registers will not be updated. In addition, the SGM41529 resets EN_ADC bit without sending any pulse on nINT pin. The ADC behaves the same if the EN_ADC bit is set while all ADC channels' enable bits are 0. To ensure that ADC is running a conversion, it is recommended to read back

EN_ADC bit after setting it to 1. The ADC conversion is interrupted if the charger mode changes during ADC conversion running (e.g. EN_OTG bit goes to 1, EN_HIZ bit goes to 1 or an adaptor plug-in). The ADC resumes with the interrupted channel when the mode change is complete.

The sample speed is programmable via ADC_SAMPLE[1:0] bits, and the ADC conversion time (t_{ADC_CONV}) for each measurement is changed according to ADC_SAMPLE[1:0] bits setting. By default, the ADC function disable register (REG0x16) is set to 0x00 and all ADC channel will be converted in both continuous and one-shot conversion modes.

If one bit of ADC function disable register (REG0x16) is set to 1 before the conversion, the corresponding channel ADC conversion data is not updated to the ADC result registers. In this case, the data host read-back from the ADC result register is from the default POR value or the last valid ADC conversion. If one bit of ADC function disable register (REG0x16) is set to 1 during the ADC conversion cycle, the SGM41529 finishes this channel conversion in this cycle, and not converts this channel in next cycle.

In continuous conversion mode, as long as the EN_ADC bit is set to 1, the ADC circuitry is active even if all ADC channel conversion is disabled by REG0x16. In one-shot mode, the EN_ADC bit is cleared automatically if all ADC channel conversion is disabled. Once one bit in REG0x16 is set to 0, the ADC conversion begins soon.

In continuous conversion mode, the ADC_DONE_STAT bit keeps 0 and ADC_DONE_FLAG bit remains unchanged. The ADC_DONE_STAT and ADC_DONE_FLAG bits are set when a one-shot conversion is completed.

To exit the ADC measurement, one of following ways is possible to do:

1. Set the EN_ADC bit to 0. The ADC measurement stops immediately. The last valid ADC measurement value can be read back into the ADC measurement result registers.
2. Set all ADC function disable bits in REG0x16 to 1. The ADC stops after the current cycle measurement is completed.
3. Set the ADC_RATE bit to 1. The ADC conversion can be set to one-shot conversion mode. The ADC stops after one cycle measurement is completed.

If an adaptor is plugged in, the ADC suspends, and it resumes after SGM41529 finishes the input source detection. Other than that, ADC conversion always keeps normal operation even a fault occurs in SGM41529.

SGM41529 I²C Controlled 2A, 2-Cell Boost Battery Charger with Power Path, OTG for USB Input

DETAILED DESCRIPTION (continued)

Status Outputs Pins (nPG, STAT and nINT) Power Good Indication (nPG Pin)

When a good input source is connected to VBUS and input type is detected, the PG_STAT status bit goes high and the nPG pin goes low. A good input source is detected if all the following conditions on V_{VBUS} are satisfied and input type detection is completed:

- V_{VBUS} is in the operating range: $V_{VBUS_UVLOZ} < V_{VBUS} < V_{VBUS_OV}$.
- Input source is not poor: $V_{VBUS} > V_{BAD_SRC} + V_{BAD_SRC_HYS}$ (100mV TYP) when I_{BAD_SRC} (15mA TYP) loading is applied. (Poor source detection.)
- Completed input source type detection.

Charge Status Indication (STAT Pin)

Charging state is indicated with the open-drain STAT pin as explained in Table 6. This pin is able to drive an LED (see Figure 1).

Table 6. STAT Pin Function

Charging State	STAT Indicator
Charging battery (or recharge)	Low (LED ON)
Charging completed	High (LED OFF)
Charging is disabled or in sleep mode	High (LED OFF)
Charge is suspended due to input over-voltage, TS fault, timer faults or battery over-voltage or OTG Buck mode is suspended (TS fault)	1Hz Blinking

nINT Interrupt Output Pin

When a new update occurs in the charger states, a 256μs negative pulse is sent through the nINT pin to interrupt the host. The host may not continuously monitor the charger device and by receiving the interrupt, it can react and check the charger situation on time. By default, each of the following events will generate a nINT pulse.

1. Good input power source is detected.
 - a) $V_{VBUS} < V_{VBUS_OV}$
 - b) $V_{VBUS} > V_{BAD_SRC}$ when I_{BAD_SRC} current is applied
2. Good input power source is removed.
3. Entering thermal regulation.
4. Entering VINDPM regulation.
5. Entering IINDPM regulation.
6. Watchdog timer expired.
7. VBUS_STAT[2:0] bits change.
8. TS_STAT[2:0] bits change.
9. CHRГ_STAT[2:0] bits change, including charge complete.
10. A rising edge on any of the *_STAT bits of REG0x0B to REG0x0E.
11. Battery over-voltage is detected.
12. Thermal shutdown.
13. VBUS over-voltage is detected.

14. Charge safety timer expired, including pre-charge timer expired.

If the event mask bit is set to 1, the corresponding nINT pulse does not send out when the event happens. For each event, there are three related bits:

1. STAT bit: holding the current status.
2. FLAG bit: holding nINT event information, ignore the current status.
3. MASK bit: prevent the event from sending out nINT.

When each of the above events occurs, the SGM41529 sends out a nINT pulse and reports the event source through the corresponding FLAG registers. After the host reads the FLAG register bits, they are automatically reset to 0, and to re-assert the FLAG requires a new rising edge on STAT bit. The example is shown in Figure 12.

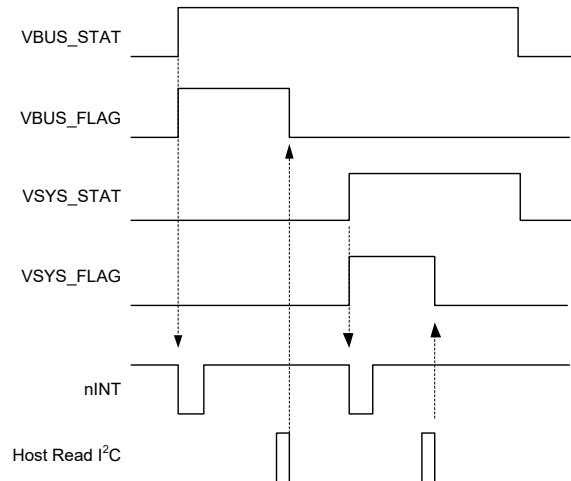


Figure 12. Example of nINT Generation Behavior

Input Current Limit on ILIM Pin

The device has an additional hardware pin on ILIM to clamp input current limit for safe operation. A resistor between ILIM pin and GND can set the clamped input current limit as:

$$I_{INMAX} = \frac{K_{ILIM}}{R_{ILIM}} \quad (3)$$

For example, if EN_ILIM = 1 and the IINDPM[4:0] register is 3A, a 750Ω resistor is connected between ILIM and ground, the actual input current limit is clamped to 1.347A ($K_{ILIM} = 1010$ TYP). When ILIM pin voltage is higher than 0.8V, the SGM41529 clamps the input current and enters input current regulation. The same behavior as entering IINDPM through IINDPM[4:0] register, the IINDPM_STAT and IINDPM_FLAG bits are also set and SGM41529 reports a nINT if the IINDPM_MASK bit is 0.

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DETAILED DESCRIPTION (continued)

When EN_ILIM = 1 and the input current is not clamped by ILIM pin, the ILIM pin voltage (V_{ILIM} , lower than 0.8V) is proportional to the actual input current. In this case, the input current can be monitored and calculated by:

$$I_{IN} = \frac{K_{ILIM} \times V_{ILIM}}{R_{ILIM} \times 0.8V} \quad (4)$$

For example, with a 750Ω ILIM resistor, the 0.4V ILIM voltage corresponds to 0.673A input current. If ILIM pin is shorted to GND, the ILIM won't help on limiting the input current and the limit is set by the IINDPM[4:0] register. If ILIM pin is open, the ILIM voltage will float above 0.8V, and the input current is limited to zero.

Setting EN_ILIM bit to 0 can disable the ILIM pin clamping function as well as the input current monitoring function. Either enable or disable ILIM pin function operation takes effect immediately.

SGM41529 Protection Features

Monitoring of Voltage and Current

During the converter operation, the input and system voltages (V_{VBUS} and V_{SYS}) and switch currents are constantly monitored to assure safe operation of the device in both Buck and Boost modes, as described below.

Boost Mode Voltage and Current Monitoring

1. Input Under-Voltage (VBUS_UVP)

The Boost converter stops switching once VBUS voltage falls below V_{BAD_SRC} during operation. During VBUS under-voltage, the PG_STAT and PG_FLAG bits can be set, and a nINT pulse is asserted to notify the host. When the under-voltage condition disappears, the device automatically resumes switching.

2. Input Over-Voltage (VBUS_OVP)

The input voltage range for Boost mode operation is above V_{BAD_SRC} and below V_{VBUS_OV} . Converter switching will stop as soon as VBUS voltage exceeds V_{VBUS_OV} over-voltage limit. During VBUS over-voltage, the SGM41529 sets the VBUS_OVP_STAT and VBUS_OVP_FLAG bits, and reports a nINT pulse if the VBUS_OVP_MASK bit is set to 0. When the over-voltage condition disappears, the Boost converter automatically resumes switching again.

3. System Over-Voltage (SYSOVP)

During a system load transient, the device clamps the system voltage to protect the system components from over-voltage. The SYSOVP threshold is related to the battery voltage.

When $V_{BAT} < V_{SYS_MIN}$, the SYSOVP threshold is $V_{SYS_MIN} + 860mV$ (TYP). When $V_{BAT} > V_{SYS_MIN}$, the SYSOVP is $V_{BAT} + 660mV$ (TYP). Once a SYSOVP occurs, switching stops to clamp any overshoot and a 12mA (TYP) sink current is applied to SYS to pull the voltage down.

4. System Over-Current (SYSOCP)

To protect system from overloading or short-circuit event, the device continually compares V_{SYS} to V_{VBUS} . If the IINDPM is triggered and the heavy load further cause the SYS voltage drop to below VBUS voltage more than 250mV, the SGM41529 stops switching and automatically tries to resume from this fault condition. If 7 consecutive retry failures are detected, the EN_HIZ bit is set automatically. The SGM41529 sets the SYS_SHORT_FLAG and reports a nINT to host if the SYS_SHORT_MASK is 0.

OTG Buck Mode Voltage and Current Monitoring

In Buck mode, the QBLK (blocking FET) and LSFET (low-side switch) FET currents and VBUS voltage are monitored for protection.

1. Output Over-Voltage Protection for VBUS

In Buck mode, converter stops switching and exits Buck mode (by clearing EN_OTG bit) if VBUS voltage rises above regulation and exceeds the V_{OTG_OVP} over-voltage limit (6V TYP). A nINT pulse is sent and the OTG_FLAG bit is set high.

2. Output Over-Current Protection for VBUS

In OTG Buck mode, when OTG output current is higher than OTG_ILIM[3:0] bits and VBUS drops to lower than 3.2V, the Buck converter stops switching. The device automatically tries to resume from this fault condition. After 7 consecutive retry failures, the EN_OTG bit is cleared, the OTG_FLAG bit is set, and a nINT is pulsed to notify the host.

Thermal Regulation and Thermal Shutdown

Boost Mode Thermal Protections

Internal junction temperature (T_J) is always monitored to avoid overheating. A limit of +120°C is considered for maximum IC surface temperature in Boost mode and if T_J intends to exceed this level, the device reduces the charge current to keep maximum temperature limited to +120°C (thermal regulation mode) and sets the TREG_STAT bit to 1. As expected, the charging current is always lower than programmed value under thermal regulation conditions. The safety timer runs at half clock rate, and the charge termination is disabled. The thermal regulation temperature is programmable from +60°C to +120°C with 20°C step.

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DETAILED DESCRIPTION (continued)

If the junction temperature exceeds T_{SHUT} (+150°C), thermal shutdown protection arises in which the converter is turned off, the SGM41529 sets the TSHUT_STAT and TSHUT_FLAG bits, and reports a nINT pulse if the TSHUT_MASK bit is set to 0. When the device recovers and T_J falls below the hysteresis band of T_{SHUT_HYS} (30°C under T_{SHUT}), the converter resumes automatically.

OTG Buck Mode Thermal Protections

Similar to Boost mode, T_J is monitored in OTG Buck mode for thermal shutdown protection. If junction temperature exceeds T_{SHUT} (+150°C), the Buck mode stops switching and the EN_OTG bit is reset to 0 automatically. Similarly, the SGM41529 sets the TSHUT_STAT and TSHUT_FLAG bits, and reports a nINT pulse if the TSHUT_MASK bit is set to 0. If T_J falls below the hysteresis band of T_{SHUT_HYS} (30°C under T_{SHUT}), the Buck mode can recover again by re-enabling EN_OTG bit by the host.

Battery Protections

Battery Over-Voltage Protection (BATOVF)

The over-voltage protection threshold of the battery pin is 4% above the programmed battery regulation voltage (V_{REG}) during charging phase. In case of a BATOVF, charging stops right away, the SGM41529 sets the BATOVF_STAT and BATOVF_FLAG bits, and reports a nINT pulse if the BATOVF_MASK bit is set to 0.

I²C Serial Interface and Data Communication

Standard I²C interface is used to program SGM41529 parameters and get status reports. I²C is the well-known 2-wire serial communication interface that can connect one (or more) master device(s) to some slave devices for two-way communication. The bus lines are named serial data (SDA) and serial clock (SCL). The device that initiates a data transfer is a master. A master generates the SCL signal. Slave devices have unique addresses to identify. A master is typically a micro controller or a digital signal processor.

The SGM41529 operates as a slave device that address is 0x6B (6BH). It has 38 8-bit registers, numbered from REG0x00 to REG0x25. A register read beyond REG0x25 (0x25) returns 0xFF.

Physical Layer

The standard I²C interface of SGM41529 supports standard mode and fast mode communication speeds. The frequency

of stand mode is up to 100kbts/s, while the fast mode is up to 400kbts/s. Bus lines are pulled high by weak current source or pull-up resistors are in logic high state with no clocking when the bus is free. The SDA pin is open-drain.

I²C Data Communication START and STOP Conditions

A transaction is started by taking control of the bus by master if the bus is free. The transaction is terminated by releasing the bus when the data transfer job is done as shown in Figure 13. All transactions are started by master which applies a START condition on the bus lines to take over the bus and exchange data. In the end, the master terminates the transaction by applying one (or more) STOP condition. START condition is generated by master when SCL is high and a high to low transition on the SDA. Similarly, a STOP is defined when SCL is high and SDA goes from low to high. After a START and before a STOP, the bus is considered busy. By the way, only a master can send out the START and STOP signals.

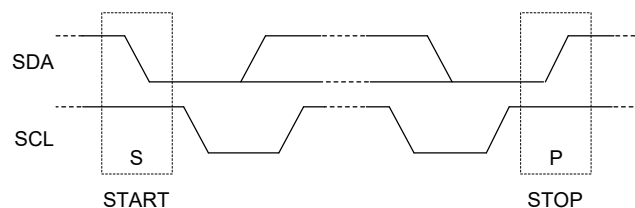


Figure 13. I²C Bus in START and STOP Conditions

Data Bit Transmission and Validity

The data bit (high or low) must remain stable on the SDA line during the high period of the clock. The state of the SDA can only change when the clock (SCL) is low. For each data bit transmission, one clock pulse is generated by master. Bit transfer in I²C is shown in Figure 14.

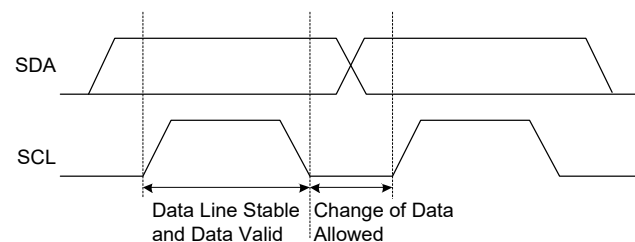


Figure 14. I²C Bus Bit Transfer

DETAILED DESCRIPTION (continued)

Byte Format

Data is transmitted in 8-bit packets (one byte at a time). The number of bytes in one transaction is not limited. In each packet, the 8 bits are sent successively with the Most Significant Bit (MSB) first. An acknowledge (or not-acknowledge) bit must come after the 8 data bits. This bit informs the transmitter whether the receiver is ready to proceed for the next byte or not. Figure 15 shows the byte transfer process with I²C interface.

Acknowledge (ACK) and Not Acknowledge (NCK)

After transmission of each byte by transmitter, an acknowledge bit is replied by the receiver as the ninth bit. With the acknowledge bit, the receiver informs the transmitter that the byte is received, and another byte is expected or can be sent (ACK) or it is not expected (NCK = not ACK). Clock (SCL) is always generated by master, including the acknowledge ninth bit, no matter who is acting as transmitter or receiver. SDA line is released for receiver control during the acknowledge clock pulse. And the receiver can pull the SDA line low as ACK (reply a 0 bit) or let it be high as NCK during the SCL high pulse. After that, the master can either

apply a STOP (P) condition to end the transaction or send a new START (S) condition to start a new transfer (called repeated start). For example, when master wants to read a register in slave, one start is needed to send the slave address and register address, and then without a STOP condition, another start is sent by master to initiate the receiving transaction from slave. Master then sends the STOP condition and releases the bus.

Data Direction Bit and Addressing Slaves

The first byte sent by master after the START is always the target slave address (7 bits) and an eighth data-direction bit (R/W). R/W bit is 0 for a WRITE transaction and 1 for READ (when master is asking for data). Data direction is the same for all next bytes of the transaction. To reverse it, a new START or repeated START condition must be sent by master (STOP will end the transaction). Usually the second byte is also a WRITE, sending the register address that is supposed to be accessed in the next byte(s). The 7-bit slave address is 1101011b (0x6B). The address bit arrangement is shown below, as shown in Figure 16 and Figure 17.

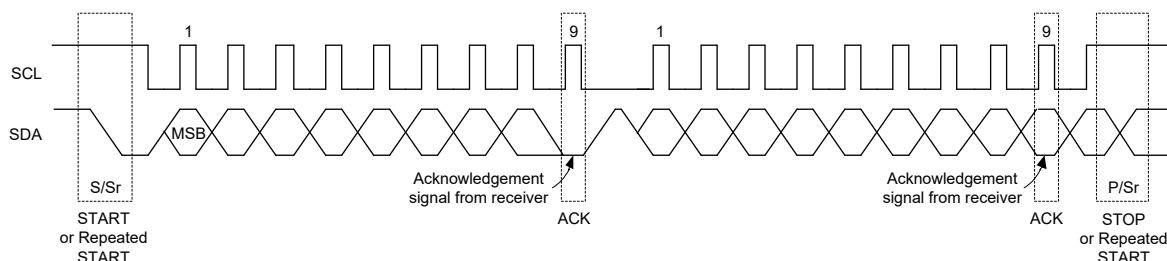


Figure 15. Byte Transfer Process

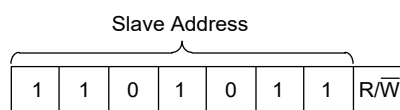


Figure 16. 7-Bit Address Format (0x6B)

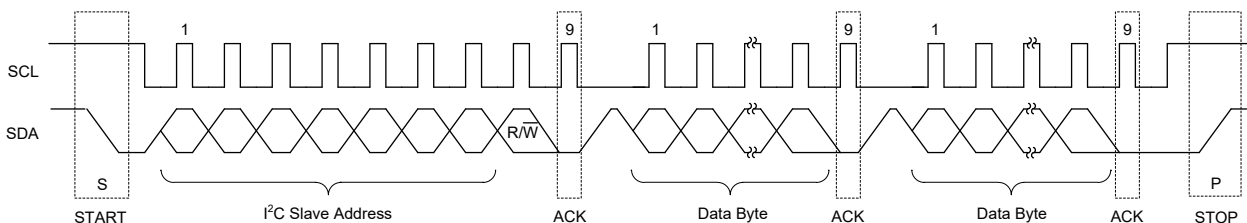


Figure 17. Data Transfer Transaction

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DETAILED DESCRIPTION (continued)

WRITE: If the master wants to write in the register, the third byte can be written directly as shown in Figure 18 for a single write data transfer. After receiving the ACK, the master may issue a STOP condition to end the transaction or send the next register data, which will be written to the next address in a slave as multi-write. A STOP is needed after sending the last data.

READ: If the master wants to read a single register (Figure 19), it sends a new START condition along with device

address with $R/\overline{W}$ bit = 1. After ACK is received, the master reads the SDA line to receive the content of the register. Master replies with NCK to inform slave that no more data is needed (single read) or it can send an ACK to request for sending the next register content (multi-read). This can continue until an NCK is sent by master. A STOP must be sent by master in any case to end the transaction.

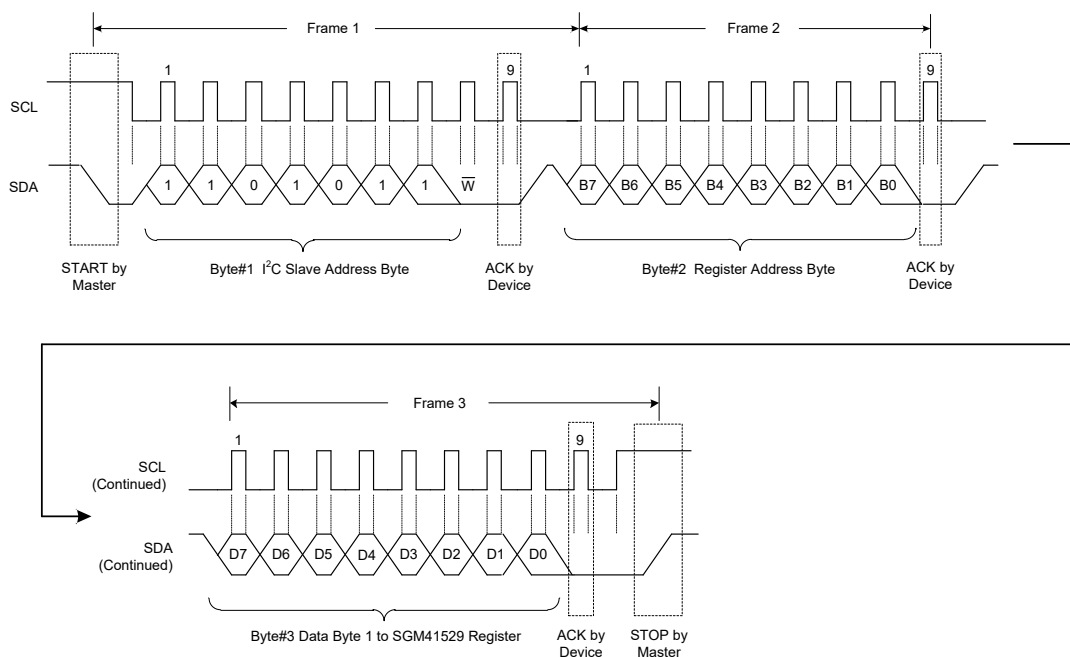


Figure 18. A Single Write Transaction

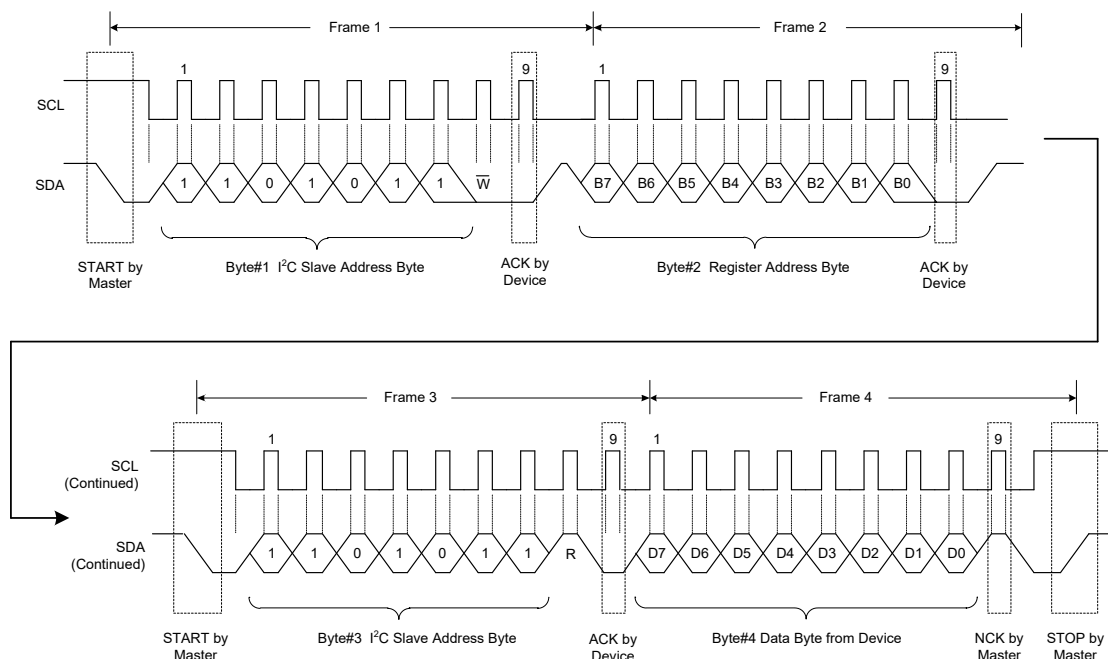


Figure 19. A Single Read Transaction

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DETAILED DESCRIPTION (continued)

Data Transactions with Multi-Read or Multi-Write

Multi-read and multi-write are supported by SGM41529 for REG0x00 through REG0x25 registers, as explained in Figure 20 and Figure 21.

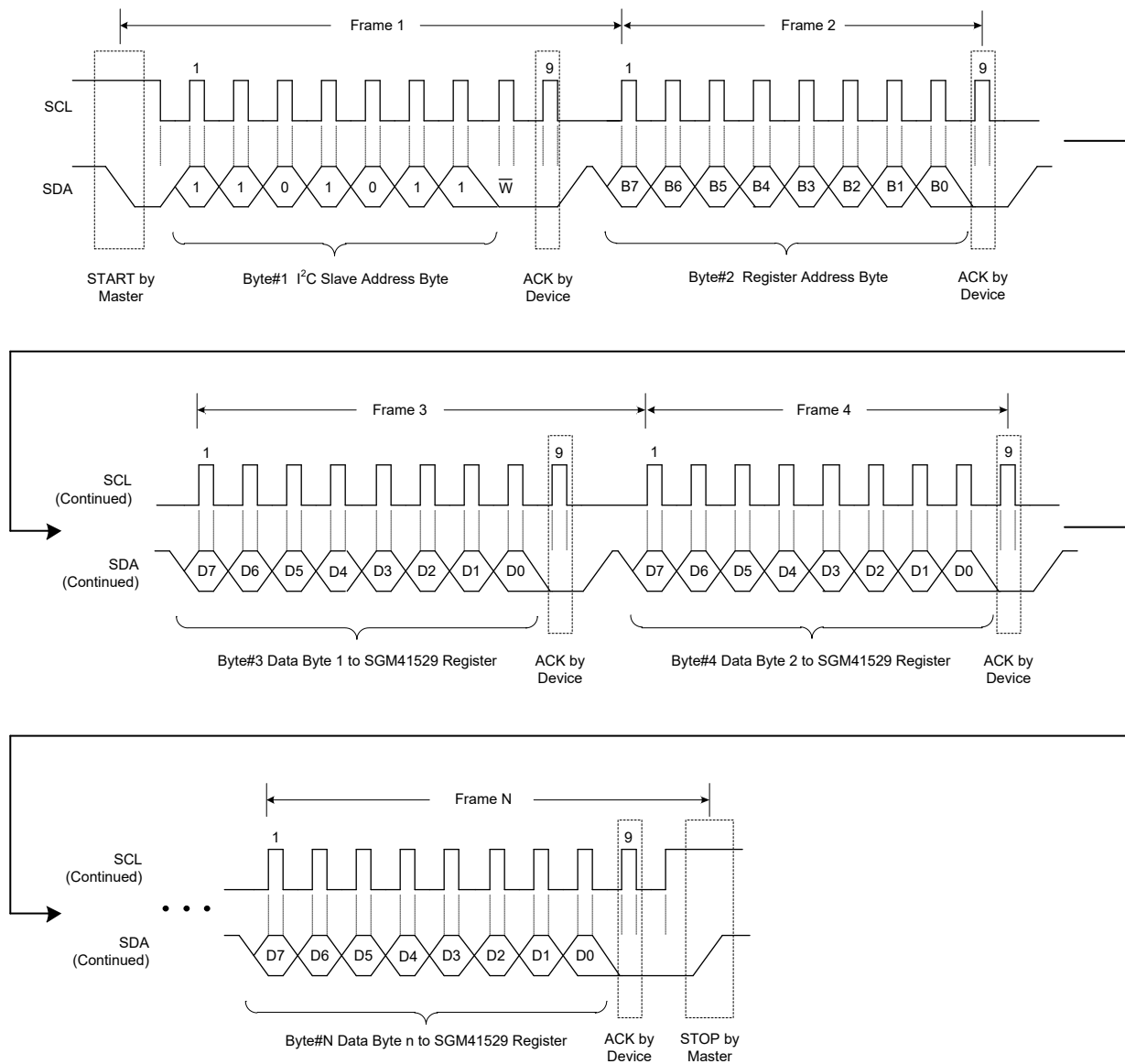


Figure 20. A Multi-Write Transaction

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DETAILED DESCRIPTION (continued)

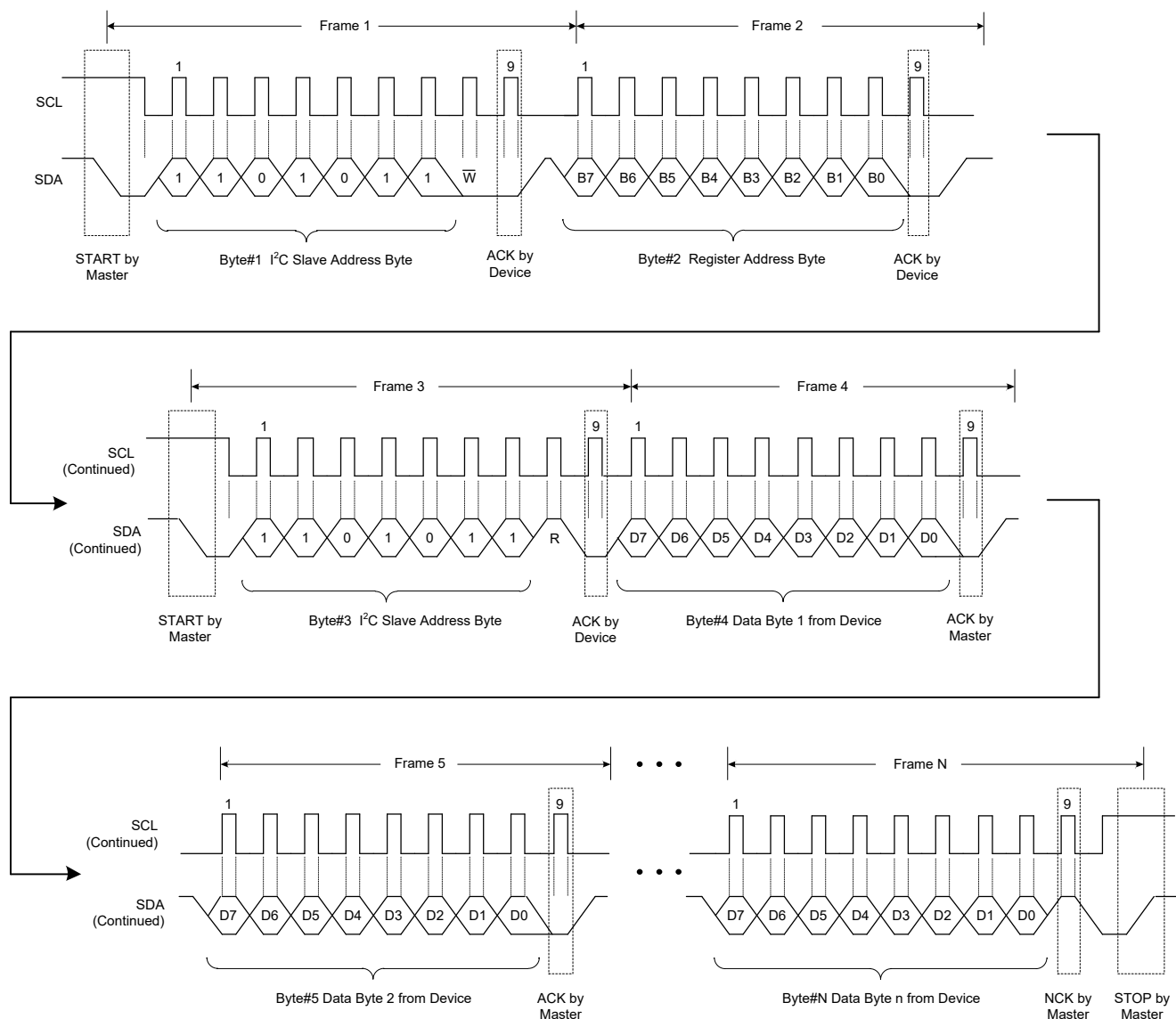


Figure 21. A Multi-Read Transaction

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REGISTER MAPS

All registers are 8-bit and individual bits are named from D[0] (LSB) to D[7] (MSB).

I²C Register Address Map

FUNCTION	STAT	FLAG	MASK	THRESHOLD SETTING	ENABLE
CHARGE	0x0B[2:0]	0x0F[0]	0x12[0]	—	0x06[3]
VREG	—	—	—	0x00[7:0]	—
ICHG	—	—	—	0x01[5:0]	—
VBATLOW	—	—	—	0x06[2]	—
IPRECHG	—	—	—	0x04[7:4]	—
TERM	—	—	—	0x04[3:0]	0x05[7]
VRECHG	—	—	—	0x06[1:0]	—
SAFETY TIMER	0x0E[4]	0x11[4]	0x14[4]	0x05[2:1]	0x05[3]
TOPOFF_TIMER	—	—	—	0x07[5:4]	0x07[5:4]
TMR2X	—	—	—	—	0x05[0]
VINDPM	0x0B[5]	0x0F[5]	0x12[5]	0x02[4:0]	—
VINDPM_RST	—	—	—	—	0x02[7]
IINDPM	0x0B[6]	0x0F[6]	0x12[6]	0x03[4:0]	—
ILIM PIN	—	—	—	—	0x01[6]
ICO	0x0C[2:1]	0x10[1]	0x13[1]	—	0x03[5]
ICO_ILIM	0x0A[4:0]	—	—	—	—
FORCE_ICO	—	—	—	—	0x03[7]
FORCE_INDET	—	—	—	—	0x03[6]
AUTO_INDET	—	—	—	—	0x06[6]
VBUS_STAT	0x0C[6:4]	0x10[4]	0x13[4]	—	—
PG	0x0C[7]	0x10[7]	0x13[7]	—	—
STAT PIN	—	—	—	—	0x05[6]
SYS_MIN	0x0C[0]	0x10[0]	0x13[0]	0x07[3:0]	—
HIZ MODE	—	—	—	—	0x01[7]
BAT_DISCHG	—	—	—	—	0x02[6]
WATCHDOG	0x0B[3]	0x0F[3]	0x12[3]	0x05[5:4]	0x05[5:4]
WD_RST	—	—	—	—	0x07[6]
OTG	0x0C[6:4]	0x11[0]	0x14[0]	—	0x06[7]
OTG_ILIM	—	—	—	0x09[7:4]	—
OTG_VLIM	—	—	—	0x09[3:0]	—
PFM	—	—	—	—	0x07[7]
PFM_OOA	—	—	—	—	0x02[5]
BHOT	—	—	—	0x08[7:6]	—
BCOLD	—	—	—	0x08[5]	—
JEITA_VSET	—	—	—	0x08[4:3]	—
JEITA_ISETH	—	—	—	0x08[2]	—
JEITA_ISETC	—	—	—	0x08[1:0]	—
TS_STAT	0x0D[2:0]	0x10[2]	0x13[2]	—	—
TREG	0x0B[4]	0x0F[4]	0x12[4]	0x06[5:4]	—
TSHUT	0x0E[6]	0x11[6]	0x14[6]	—	—
VBUS_OVP	0x0E[7]	0x11[7]	0x14[7]	—	—
BATOVP	0x0E[5]	0x11[5]	0x14[5]	—	—
SYS_SHORT	—	0x11[3]	0x14[3]	—	—
ADC	0x0B[7]	0x0F[7]	0x12[7]	—	0x15[7]
ADC_RATE	—	—	—	0x15[6]	—
ADC_SAMPLE	—	—	—	0x15[5:4]	—
REG_RST	—	—	—	—	0x25[7]

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REGISTER MAPS (continued)

Bit Types:

R: Read only

R/W: Read/Write

RC: Read clears the bit

R/WC: Read/Write. Writing a '1' clears the bit. Writing a '0' has no effect.

REG0x00: Battery Voltage Regulation Limit Register [Reset = 0xA0]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:0]	VREG[7:0]	1010 0000	R/W	0000 0000 = 6.8V 0000 0001 = 6.81V 0000 0010 = 6.82V 1010 0000 = 8.4V (default) 1011 1110 = 8.7V 1100 1000 = 8.8V 1110 1110 = 9.18V 1110 1111 = 9.19V 1111 0000 = 9.2V Note: V _{REG} setting range is 6.8V to 9.2V. V _{REG} > 9.2V clamped at 9.2V.	REG_RST or Watchdog

REG0x01: Charge Current Limit Register [Reset = 0x5E]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	EN_HIZ	0	R/W	0 = Disable HIZ mode (default) 1 = Enable HIZ mode	REG_RST or Watchdog
D[6]	EN_ILIM	1	R/W	0 = Disable ILIM pin function 1 = Enable ILIM pin function (default)	REG_RST or Watchdog
D[5:0]	ICRG[5:0]	01 1110	R/W	00 0010 = 100mA 00 0011 = 150mA 00 0100 = 200mA 01 1110 = 1500mA (default) 10 1010 = 2100mA 10 1011 = 2150mA 10 1100 = 2200mA Note: I _{CHG} setting range is 100mA to 2200mA. I _{CHG} > 2.2A clamped at 2.2A. I _{CHG} < 100mA clamped at 100mA.	REG_RST or Watchdog

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REGISTER MAPS (continued)

REG0x02: Input Voltage Limit Register [Reset = 0x84]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	EN_VINDPM_RST	1	R/W	Enable VINDPM Reset to Default upon Adaptor Plug-In 0 = Disable VINDPM reset when adaptor is plugged in 1 = Enable VINDPM reset when adaptor is plugged in (VINDPM resets to default value) (default)	REG_RST or Watchdog
D[6]	EN_BAT_DISCHG	0	R/W	Enable BAT Pin Discharge Load (I _{BAT_DISCHG}) 0 = Disable BAT discharge load (default) 1 = Enable BAT discharge load	REG_RST or Watchdog
D[5]	PFM_OOA_DIS	0	R/W	PFM Out-of-Audio (OOA) Mode Disable 0 = Out-of-audio mode enabled while converter is in PFM (default) 1 = Out-of-audio mode disabled while converter is in PFM	REG_RST
D[4:0]	VINDPM[4:0]	0 0100	R/W	VINDPM Threshold Setting Bits 0 0000 = 3.9V 0 0001 = 4.0V 0 0010 = 4.1V 0 0100 = 4.3V (default) 0 1110 = 5.3V 0 1111 = 5.4V 1 0000 = 5.5V Note: V _{INDPM} setting range is 3.9V to 5.5V. V _{INDPM} > 5.5V clamped at 5.5V.	REG_RST

REG0x03: Input Current Limit Register [Reset = 0x39]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	FORCE_ICO	0	R/W	Force Start Input Current Optimizer (ICO) 0 = Do not force ICO (default) 1 = Force ICO start Note: The bit can only be set and automatically returns 0 after ICO starts. The bit only valid when EN_ICO = 1.	REG_RST or Watchdog
D[6]	FORCE_INDET	0	R/W	Force D+/D- Detection 0 = Not in D+/D- detection (default) 1 = Force D+/D- detection	REG_RST or Watchdog
D[5]	EN_ICO	1	R/W	Input Current Optimization (ICO) Algorithm Control 0 = Disable ICO 1 = Enable ICO (default)	REG_RST
D[4:0]	IINDPM[4:0]	1 1001	R/W	IINDPM Threshold Setting Bits 0 0000 = 500mA 0 0001 = 600mA 0 0010 = 700mA 1 1001 = 3000mA (default) 1 1010 = 3100mA 1 1011 = 3200mA 1 1100 = 3300mA Note: I _{INDPM} setting range is 0.5A to 3.3A. I _{INDPM} > 3.3A clamped at 3.3A.	REG_RST

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REGISTER MAPS (continued)

REG0x04: Pre-Charge and Termination Current Limit Register [Reset = 0x22]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:4]	IPRECHG[3:0]	0010	R/W	Pre-Charge Current Limit 0000 = 50mA 0001 = 100mA 0010 = 150mA (default) 1101 = 700mA 1110 = 750mA 1111 = 800mA	REG_RST or Watchdog
D[3:0]	ITERM[3:0]	0010	R/W	Termination Current Limit 0000 = 50mA 0001 = 100mA 0010 = 150mA (default) 1101 = 700mA 1110 = 750mA 1111 = 800mA	REG_RST or Watchdog

REG0x05: Charger Control 1 Register [Reset = 0x9D]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	EN_TERM	1	R/W	Charging Termination Enable 0 = Disable termination 1 = Enable termination (default)	REG_RST or Watchdog
D[6]	STAT_DIS	0	R/W	STAT Pin Disable 0 = Enable STAT pin function (default) 1 = Disable STAT pin function	REG_RST or Watchdog
D[5:4]	WATCHDOG[1:0]	01	R/W	I ² C Watchdog Timer Setting 00 = Disable watchdog timer 01 = 40s (default) 10 = 80s 11 = 160s Note: The watchdog timer will not restart if modifying the WATCHDOG[1:0] register before the watchdog timer expires.	REG_RST or Watchdog
D[3]	EN_TIMER	1	R/W	Charge Safety Timer Enable 0 = Disable 1 = Enable (default)	REG_RST or Watchdog
D[2:1]	CHG_TIMER[1:0]	10	R/W	Fast Charge Timer Setting 00 = 7.5h 01 = 12h 10 = 16.5h (default) 11 = 21h	REG_RST or Watchdog
D[0]	EN_TMR2X	1	R/W	Enable Half Clock Rate Safety Timer 0 = Disable 1 = Safety timer slow down (by a factor of 2) during DPM or thermal regulation (default)	REG_RST or Watchdog

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REGISTER MAPS (continued)

REG0x06: Charger Control 2 Register [Reset = 0x7D]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	EN_OTG	0	R/W	Buck (OTG) Mode Control 0 = Disable OTG (default) 1 = Enable OTG	REG_RST or Watchdog
D[6]	EN_AUTO_INDET	1	R/W	Automatic D+/D- Detection Enable 0 = Disable D+/D- detection when VBUS plugs in 1 = Enable D+/D- detection when VBUS plugs in (default)	REG_RST or Watchdog
D[5:4]	TREG[1:0]	11	R/W	Thermal Regulation Threshold 00 = +60°C 01 = +80°C 10 = +100°C 11 = +120°C (default)	REG_RST or Watchdog
D[3]	EN_CHG	1	R/W	Charger Enable Configuration 0 = Charge disable 1 = Charge enable (default)	REG_RST or Watchdog
D[2]	VBATLOW	1	R/W	Battery Pre-Charge to Fast Charge Threshold 0 = 5.6V 1 = 6V (default)	REG_RST or Watchdog
D[1:0]	VRECHG[1:0]	01	R/W	Battery Recharge Threshold Setting (below V _{REG}) 00 = 100mV 01 = 200mV (default) 10 = 300mV 11 = 400mV	REG_RST

REG0x07: Charger Control 3 Register [Reset = 0x02]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	PFM_DIS	0	R/W	PFM Mode Disable 0 = Enable PFM operation (default) 1 = Disable PFM operation	REG_RST
D[6]	WD_RST	0	R/W	I ² C Watchdog Timer Reset 0 = Normal (default) 1 = Reset (bit goes back to 0 after timer reset)	REG_RST or Watchdog
D[5:4]	TOPOFF_TIMER[1:0]	00	R/W	Top-Off Timer 00 = Disabled (default) 01 = 15min 10 = 30min 11 = 45min	REG_RST or Watchdog
D[3:0]	SYS_MIN[3:0]	0010	R/W	Minimum System Voltage Limit 0000 = 6.0V 0001 = 6.1V 0010 = 6.2V (default) 1010 = 7.0V 1101 = 7.3V 1110 = 7.4V 1111 = 7.5V	REG_RST

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REGISTER MAPS (continued)

REG0x08: Charger Control 4 Register [Reset = 0x0D]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:6]	BHOT[1:0]	00	R/W	TS Hot Temperature Threshold in OTG Mode 00 = V _{BHOT1} threshold (34.75%) (default) 01 = V _{BHOT0} threshold (37.75%) 10 = V _{BHOT2} threshold (31.25%) 11 = Disable OTG mode thermal protection	REG_RST or Watchdog
D[5]	BCOLD	0	R/W	TS Cold Temperature Threshold in OTG Mode 0 = V _{BCOLD0} threshold (77%) (default) 1 = V _{BCOLD1} threshold (80%)	REG_RST or Watchdog
D[4:3]	JEITA_VSET[1:0] (45°C - 60°C)	01	R/W	JEITA High Temperature Voltage Setting 00 = Charge suspend 01 = Set V _{REG} to 8V (default) 10 = Set V _{REG} to 8.3V 11 = V _{REG} unchanged	REG_RST or Watchdog
D[2]	JEITA_ISETH (45°C - 60°C)	1	R/W	JEITA High Temperature Current Setting 0 = 40% of I _{CHG} 1 = 100% of I _{CHG} (default)	REG_RST or Watchdog
D[1:0]	JEITA_ISETC[1:0] (0°C - 10°C)	01	R/W	JEITA Low Temperature Current Setting 00 = Charge suspend 01 = 20% of I _{CHG} (default) 10 = 40% of I _{CHG} 11 = 100% of I _{CHG}	REG_RST or Watchdog

REG0x09: OTG Control Register [Reset = 0xF6]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:4]	OTG_ILIM[3:0]	1111	R/W	Buck (OTG) Mode Current Limit 0000 = 0.5A 0001 = 0.6A 0010 = 0.7A 1101 = 1.8A 1110 = 1.9A 1111 = 2.0A (default)	REG_RST or Watchdog
D[3:0]	OTG_VLIM[3:0]	0110	R/W	Buck (OTG) Mode Regulation Voltage 0000 = 4.5V 0001 = 4.6V 0010 = 4.7V 0110 = 5.1V (default) 0111 = 5.2V 1000 = 5.3V 1001 = 5.4V 1010 = 5.5V Note: OTG_VLIM[3:0] setting range is 4.5V to 5.5V. OTG_VLIM[3:0] > 5.5V clamped at 5.5V.	REG_RST or Watchdog

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REGISTER MAPS (continued)

REG0x0A: ICO Current Limit Register [Reset = 0xXX]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:5]	Reserved	000	R	Reserved.	N/A
D[4:0]	ICO_ILIM[4:0]	x xxxx	R	Actual Input Current Limit 0 0000 = 500mA 0 0001 = 600mA 0 0010 = 700mA 1 1010 = 3100mA 1 1011 = 3200mA 1 1100 = 3300mA Note: The maximum input current limit is 3300mA.	N/A

REG0x0B: Charger Status 1 Register [Reset = 0xXX]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	ADC_DONE_STAT	x	R	One-Shot Mode ADC Conversion Status 0 = Conversion is not completed 1 = Conversion is completed Note: Always reads 0 in continuous mode	N/A
D[6]	IINDPM_STAT	x	R	Input Current Regulation (Dynamic Power Management) 0 = Not in IINDPM regulation 1 = In IINDPM regulation (ILIM pin or IINDPM[4:0] register)	N/A
D[5]	VINDPM_STAT	x	R	Input Voltage Regulation (Dynamic Power Management) 0 = Not in VINDPM regulation 1 = In VINDPM regulation	N/A
D[4]	TREG_STAT	x	R	IC Thermal Regulation Status 0 = Normal 1 = In thermal regulation	N/A
D[3]	WD_STAT	x	R	I ² C Watchdog Timer Status Bit 0 = Normal 1 = Watchdog timer expired	N/A
D[2:0]	CHRG_STAT[2:0]	xxx	R	Charge Status Bits 000 = Not charging 001 = Trickle charge ($V_{BAT} < V_{BAT_SHORT}$) 010 = Pre-charge ($V_{BAT_SHORT} < V_{BAT} < V_{BAT_LOW}$) 011 = Fast charge (CC mode) 100 = Taper charge (CV mode) 101 = Top-off timer active charging 110 = Charge termination done 111 = Reserved	N/A

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REGISTER MAPS (continued)

REG0x0C: Charger Status 2 Register [Reset = 0xXX]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	PG_STAT	x	R	Input Power Status (VBUS in Good Voltage Range and Not Poor) 0 = Input power source is not good 1 = Input power source is good	N/A
D[6:4]	VBUS_STAT[2:0]	xxx	R	VBUS Status Register 000 = No input 001 = USB host SDP (500mA) 010 = USB CDP (1.5A) 011 = USB DCP (3.0A) 100 = POORSRC detected 7 consecutive times 101 = Unknown adaptor (500mA) 110 = Non-standard adaptor (1A/2.1A/2.4A) 111 = OTG	N/A
D[3]	Reserved	0	R	Reserved.	N/A
D[2:1]	ICO_STAT[1:0]	xx	R	Input Current Optimizer (ICO) Status 00 = ICO disabled 01 = ICO optimization is in progress 10 = Maximum input current detected 11 = Reserved	N/A
D[0]	VSYS_STAT	x	R	System Voltage Regulation Status 0 = Not in SYS_MIN regulation ($V_{BAT} > V_{SYS_MIN}$) 1 = In SYS_MIN regulation ($V_{BAT} < V_{SYS_MIN}$)	N/A

REG0x0D: NTC Status Register [Reset = 0x0X]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:3]	Reserved	0 0000	R	Reserved.	N/A
D[2:0]	TS_STAT[2:0]	xxx	R	NTC (TS) Status 000 = Normal 010 = TS warm 011 = TS cool 101 = TS cold 110 = TS hot	N/A

REG0x0E: FAULT Status Register [Reset = 0xX0]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	VBUS_OVP_STAT	x	R	Input Over-Voltage Status 0 = Normal 1 = Device in VBUS over-voltage protection	N/A
D[6]	TSHUT_STAT	x	R	Temperature Shutdown Status 0 = Normal 1 = Device in thermal shutdown protection	N/A
D[5]	BATOVP_STAT	x	R	Battery Over-Voltage Status 0 = Normal 1 = BATOVP ($V_{BAT} > V_{BATOVP}$)	N/A
D[4]	TMR_STAT	x	R	Charge Safety Timer Status 0 = Normal 1 = Charge safety timer expired	N/A
D[3:0]	Reserved	0000	R	Reserved.	N/A

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REGISTER MAPS (continued)

REG0x0F: Charger Flag 1 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	ADC_DONE_FLAG	0	R	One-Shot Mode ADC Conversion Flag 0 = Conversion is not completed (default) 1 = Conversion is completed Note: Always reads 0 in continuous mode.	REG_RST
D[6]	IINDPM_FLAG	0	RC	IINDPM Regulation nINT Flag 0 = Normal (default) 1 = IINDPM signal rising edge is detected	REG_RST
D[5]	VINDPM_FLAG	0	RC	VINDPM Regulation nINT Flag 0 = Normal (default) 1 = VINDPM signal rising edge is detected	REG_RST
D[4]	TREG_FLAG	0	RC	IC Temperature Regulation nINT Flag 0 = Normal (default) 1 = TREG signal rising edge is detected	REG_RST
D[3]	WD_FLAG	0	RC	I ² C Watchdog nINT Flag 0 = Normal (default) 1 = WD_STAT signal rising edge is detected	REG_RST
D[2:1]	Reserved	00	R	Reserved.	N/A
D[0]	CHRG_FLAG	0	RC	Charge Status nINT Flag 0 = Normal (default) 1 = CHRG_STAT[2:0] bits are changed	REG_RST

REG0x10: Charger Flag 2 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	PG_FLAG	0	RC	Power Good nINT Flag 0 = Normal (default) 1 = PG signal toggle detected	REG_RST
D[6:5]	Reserved	00	R	Reserved.	N/A
D[4]	VBUS_FLAG	0	RC	VBUS Status nINT Flag 0 = Normal (default) 1 = VBUS_STAT[2:0] bits are changed	REG_RST
D[3]	Reserved	0	R	Reserved.	N/A
D[2]	TS_FLAG	0	RC	TS Status nINT Flag 0 = Normal (default) 1 = TS_STAT[2:0] bits are changed	REG_RST
D[1]	ICO_FLAG	0	RC	Input Current Optimizer (ICO) nINT Flag 0 = Normal (default) 1 = ICO_STAT[1:0] bits are changed	REG_RST
D[0]	VSYS_FLAG	0	RC	System Voltage Regulation nINT Flag 0 = Normal (default) 1 = Enter or exit SYS_MIN regulation	REG_RST

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REGISTER MAPS (continued)

REG0x11: FAULT Flag Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	VBUS_OVP_FLAG	0	RC	Input Over-Voltage nINT Flag 0 = Normal (default) 1 = Enter VBUS_OVP fault	REG_RST
D[6]	TSHUT_FLAG	0	RC	IC Temperature Shutdown nINT Flag 0 = Normal (default) 1 = Enter TSHUT fault	REG_RST
D[5]	BATOVP_FLAG	0	RC	Battery Over-Voltage nINT Flag 0 = Normal (default) 1 = Enter BATOVP fault	REG_RST
D[4]	TMR_FLAG	0	RC	Charge Safety Timer Fault nINT Flag 0 = Normal (default) 1 = Charge safety timer expired rising edge is detected	REG_RST
D[3]	SYS_SHORT_FLAG	0	RC	System Short nINT Flag 0 = Normal (default) 1 = Stop switching due to Boost converter overload	REG_RST
D[2:1]	Reserved	00	R	Reserved.	N/A
D[0]	OTG_FLAG	0	RC	OTG Buck Mode Fault nINT Flag 0 = Normal (default) 1 = VBUS overload in OTG, or VBUS_OVP, or $V_{BAT} < V_{OTG_BAT}$	REG_RST

REG0x12: Charger Mask 1 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	ADC_DONE_MASK	0	R/W	ADC Conversion nINT Mask Flag 0 = ADC_DONE produces nINT pulse (default) 1 = ADC_DONE does not produce nINT pulse Note: Only one-shot mode.	REG_RST
D[6]	IINDPM_MASK	0	R/W	IINDPM Regulation nINT Mask 0 = IINDPM entry produces nINT pulse (default) 1 = IINDPM entry does not produce nINT pulse	REG_RST
D[5]	VINDPM_MASK	0	R/W	VINDPM Regulation nINT Mask 0 = VINDPM entry produces nINT pulse (default) 1 = VINDPM entry does not produce nINT pulse	REG_RST
D[4]	TREG_MASK	0	R/W	IC Temperature Regulation nINT Mask 0 = TREG entry produces nINT pulse (default) 1 = TREG entry does not produce nINT pulse	REG_RST
D[3]	WD_MASK	0	R/W	I ² C Watchdog Timer nINT Mask 0 = WD_STAT bit rising edge produces nINT pulse (default) 1 = WD_STAT bit rising edge does not produce nINT pulse	REG_RST
D[2:1]	Reserved	00	R	Reserved.	N/A
D[0]	CHRG_MASK	0	R/W	Charge Status nINT Mask 0 = CHRG_STAT[2:0] bits change produces nINT pulse (default) 1 = CHRG_STAT[2:0] bits change does not produce nINT pulse	REG_RST

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REGISTER MAPS (continued)

REG0x13: Charger Mask 2 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	PG_MASK	0	R/W	Power Good nINT Mask 0 = PG toggle produces nINT pulse (default) 1 = PG toggle does not produce nINT pulse	REG_RST
D[6:5]	Reserved	00	R	Reserved.	N/A
D[4]	VBUS_MASK	0	R/W	VBUS Status nINT Mask 0 = VBUS_STAT[2:0] bits change produces nINT pulse (default) 1 = VBUS_STAT[2:0] bits change does not produce nINT pulse	REG_RST
D[3]	Reserved	0	R	Reserved.	N/A
D[2]	TS_MASK	0	R/W	TS Status nINT Mask 0 = TS_STAT[2:0] bits change produces nINT pulse (default) 1 = TS_STAT[2:0] bits change does not produce nINT pulse	REG_RST
D[1]	ICO_MASK	0	R/W	Input Current Optimizer (ICO) nINT Mask 0 = ICO_STAT rising edge produces nINT pulse (default) 1 = ICO_STAT rising edge does not produce nINT pulse	REG_RST
D[0]	VSYS_MASK	0	R/W	System Voltage Regulation nINT Mask 0 = Entering or exiting SYS_MIN produces nINT pulse (default) 1 = Entering or exiting SYS_MIN does not produce nINT pulse	REG_RST

REG0x14: Fault Mask Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	VBUS_OVP_MASK	0	R/W	Input Over-Voltage nINT Mask 0 = VBUS_OVP rising edge produces nINT pulse (default) 1 = VBUS_OVP rising edge does not produce nINT pulse	REG_RST
D[6]	TSHUT_MASK	0	R/W	Thermal Shutdown nINT Mask 0 = TSHUT rising edge produces nINT pulse (default) 1 = TSHUT rising edge does not produce nINT pulse	REG_RST
D[5]	BATOVP_MASK	0	R/W	Battery Over-Voltage nINT Mask 0 = BATOVP rising edge produces nINT pulse (default) 1 = BATOVP rising edge does not produce nINT pulse	REG_RST
D[4]	TMR_MASK	0	R/W	Charge Safety Timer Fault nINT Mask 0 = Timer expired rising edge produces nINT pulse (default) 1 = Timer expired rising edge does not produce nINT pulse	REG_RST
D[3]	SYS_SHORT_MASK	0	R/W	System Short Fault nINT Mask 0 = System short rising edge produces nINT pulse (default) 1 = System short rising edge does not produce nINT pulse	REG_RST
D[2:1]	Reserved	00	R	Reserved.	N/A
D[0]	OTG_MASK	0	R/W	OTG Buck Mode Fault nINT Mask 0 = OTG Buck mode fault event produces nINT pulse (default) 1 = OTG Buck mode fault event does not produce nINT pulse	REG_RST

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REGISTER MAPS (continued)

REG0x15: ADC Control Register [Reset = 0x30]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	EN_ADC	0	R/W	ADC Control 0 = Disable ADC (default) 1 = Enable ADC	REG_RST or Watchdog
D[6]	ADC_RATE	0	R/W	0 = Continuous conversion (default) 1 = One-shot conversion	REG_RST
D[5:4]	ADC_SAMPLE[1:0]	11	R/W	Sample Speed of ADC 00 = 15-bit effective resolution 01 = 14-bit effective resolution 10 = 13-bit effective resolution 11 = 12-bit effective resolution (default)	REG_RST
D[3:0]	Reserved	0000	R	Reserved.	N/A

REG0x16: ADC Function Disable Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	IBUS_ADC_DIS	0	R/W	0 = Enable conversion (default) 1 = Disable conversion	REG_RST
D[6]	ICHG_ADC_DIS	0	R/W	0 = Enable conversion (default) 1 = Disable conversion	REG_RST
D[5]	VBUS_ADC_DIS	0	R/W	0 = Enable conversion (default) 1 = Disable conversion	REG_RST
D[4]	VBAT_ADC_DIS	0	R/W	0 = Enable conversion (default) 1 = Disable conversion	REG_RST
D[3]	VSYS_ADC_DIS	0	R/W	0 = Enable conversion (default) 1 = Disable conversion	REG_RST
D[2]	TS_ADC_DIS	0	R/W	0 = Enable conversion (default) 1 = Disable conversion	REG_RST
D[1]	Reserved	0	R	Reserved.	N/A
D[0]	TDIE_ADC_DIS	0	R/W	0 = Enable conversion (default) 1 = Disable conversion	REG_RST

REG0x17: IBUS ADC 1 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	IBUS_POL	0	R	Polarity of IBUS 0 = Positive (positive means flow into VBUS pin) (default) 1 = Negative (negative means flow out of VBUS pin)	REG_RST
D[6:4]	Reserved	000	R	Reserved. Note: The reserved bits REG0x17[6:4] need to be skipped when calculating the decimal current. For example, the result 8C18h in IBUS_ADC represents -1000mA IBUS current (OTG output current).	N/A
D[3:0]	IBUS_ADC[11:8]	0000	R	Higher 4 Bits of the 12-Bit ADC IBUS Data (1mA Resolution) MSB<3:0>: 2048mA, 1024mA, 512mA, 256mA Note: IBUS reading range: 0A - 4A.	REG_RST

REG0x18: IBUS ADC 0 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:0]	IBUS_ADC[7:0]	0000 0000	R	Lower Bits of the 12-Bit ADC IBUS Data (1mA Resolution) LSB<7:0>: 128mA, 64mA, 32mA, 16mA, 8mA, 4mA, 2mA, 1mA Note: IBUS reading range: 0A - 4A.	REG_RST

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REGISTER MAPS (continued)

REG0x19: ICHG ADC 1 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:4]	Reserved	0000	R	Reserved.	N/A
D[3:0]	ICHG_ADC[11:8]	0000	R	Higher 4 Bits of the 12-Bit ADC ICHG Data (1mA Resolution) MSB<3:0>: 2048mA, 1024mA, 512mA, 256mA Note: ICHG reading range: 0A - 4A.	REG_RST

REG0x1A: ICHG ADC 0 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:0]	ICHG_ADC[7:0]	0000 0000	R	Lower Bits of the 12-Bit ADC ICHG Data (1mA Resolution) LSB<7:0>: 128mA, 64mA, 32mA, 16mA, 8mA, 4mA, 2mA, 1mA Note: ICHG reading range: 0A - 4A.	REG_RST

REG0x1B: VBUS ADC 1 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:5]	Reserved	000	R	Reserved.	N/A
D[4:0]	VBUS_ADC[12:8]	0 0000	R	Higher 5 Bits of the 13-Bit ADC VBUS Data (1mV Resolution) MSB<4:0>: 4096mV, 2048mV, 1024mV, 512mV, 256mV Note: VBUS reading range: 0V - 6.5V. And EN_ADC bit is cleared when $V_{VBUS} > V_{VBUS_OV}$.	REG_RST

REG0x1C: VBUS ADC 0 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:0]	VBUS_ADC[7:0]	0000 0000	R	Lower Bits of the 13-Bit ADC VBUS Data (1mV Resolution) LSB<7:0>: 128mV, 64mV, 32mV, 16mV, 8mV, 4mV, 2mV, 1mV Note: VBUS reading range: 0V - 6.5V. And EN_ADC bit is cleared when $V_{VBUS} > V_{VBUS_OV}$.	REG_RST

REG0x1D: VBAT ADC 1 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:6]	Reserved	00	R	Reserved.	N/A
D[5:0]	VBAT_ADC[13:8]	00 0000	R	Higher 6 Bits of the 14-Bit ADC VBAT Data (1mV Resolution) MSB<5:0>: 8192mV, 4096mV, 2048mV, 1024mV, 512mV, 256mV Note: VBAT reading range: 0V - 10V.	REG_RST

REG0x1E: VBAT ADC 0 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:0]	VBAT_ADC[7:0]	0000 0000	R	Lower Bits of the 14-Bit ADC VBAT Data (1mV Resolution) LSB<7:0>: 128mV, 64mV, 32mV, 16mV, 8mV, 4mV, 2mV, 1mV Note: VBAT reading range: 0V - 10V.	REG_RST

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REGISTER MAPS (continued)

REG0x1F: VSYS ADC 1 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:6]	Reserved	00	R	Reserved.	N/A
D[5:0]	VSYS_ADC[13:8]	00 0000	R	Higher 6 Bits of the 14-Bit ADC VSYS Data (1mV Resolution) MSB<5:0>: 8192mV, 4096mV, 2048mV, 1024mV, 512mV, 256mV Note: VSYS reading range: 0V - 10V.	REG_RST

REG0x20: VSYS ADC 0 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:0]	VSYS_ADC[7:0]	0000 0000	R	Lower Bits of the 14-Bit ADC VSYS Data (1mV Resolution) LSB<7:0>: 128mV, 64mV, 32mV, 16mV, 8mV, 4mV, 2mV, 1mV Note: VSYS reading range: 0V - 10V.	REG_RST

REG0x21: TS ADC 1 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:2]	Reserved	00 0000	R	Reserved.	N/A
D[1:0]	TS_ADC[9:8]	00	R	Higher 2 Bits of the 10-Bit ADC TS Data (0.098% Resolution) MSB<1:0>: 50%, 25% Note: TS as percentage of REGN valid reading range: 10% - 90%. Accuracy is not guaranteed beyond this range.	REG_RST

REG0x22: TS ADC 0 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:0]	TS_ADC[7:0]	0000 0000	R	Lower Bits of the 10-Bit ADC TS Data (0.098% Resolution) LSB<7:0>: 12.5%, 6.25%, 3.125%, 1.563%, 0.781%, 0.391%, 0.195%, 0.098% Note: TS as percentage of REGN valid reading range: 10% - 90%. Accuracy is not guaranteed beyond this range.	REG_RST

REG0x23: TDIE ADC 1 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:1]	Reserved	000 0000	R	Reserved.	N/A
D[0]	TDIE_ADC[8]	0	R	Higher 1 Bit of the 9-Bit ADC TDIE Data (0.5°C Resolution) MSB<0>: 128°C Note: TDIE (IC temperature) reading range: 0°C - 128°C.	REG_RST

REG0x24: TDIE ADC 0 Register [Reset = 0x00]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7:0]	TDIE_ADC[7:0]	0000 0000	R	Lower Bits of the 9-Bit ADC TDIE Data (0.5°C Resolution) LSB<7:0>: 64°C, 32°C, 16°C, 8°C, 4°C, 2°C, 1°C, 0.5°C Note: TDIE (IC temperature) reading range: 0°C - 128°C.	REG_RST

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REGISTER MAPS (continued)

REG0x25: Part Information Register [Reset = 0x18]

BITS	BIT NAME	DEFAULT	TYPE	DESCRIPTION	RESET BY
D[7]	REG_RST	0	R/WC	Register Reset 0 = No effect (keep current register settings) (default) 1 = Reset R/W bits of all registers to the default and reset safety timer (it also resets itself to 0 after register reset is completed)	REG_RST
D[6:3]	PN[3:0]	0011	R	Part ID 0011 = SGM41529	N/A
D[2:0]	DEV_REV[2:0]	001	R	Revision: 001	N/A

APPLICATION INFORMATION

The SGM41529 is typically used as a charger with power path management in smart phones, tablets and other portable devices. In the design, it comes along with a host controller (a processor with I²C interface) and a 2-cell Li-Ion or Li-polymer battery.

Detailed Design Procedure
Inductor Selection

The inductor selection mainly considers the inductance, the saturation current and heat rating current. The saturation current and heat rating current are better to higher than the possible maximum current considering the inductor current ripple as following formula:

$$I_{SAT} > I_{IN} + \frac{\Delta I}{2} \quad (5)$$

The inductor ripple current (ΔI) depends on input voltage (V_{VBUS}), output voltage (V_{SYS}), inductor (L) and switching frequency (f_{SW}):

$$\Delta I = \frac{V_{VBUS} \times (V_{SYS} - V_{VBUS})}{V_{SYS} \times f_{SW} \times L} \quad (6)$$

An inductor with a larger value results in less ripple current and a lower peak inductor current, reducing stress on the power MOSFET. However, the larger value inductor has a larger physical size, a higher series resistance, and a lower saturation current. For trade-off between the inductor power loss and size, it is recommended to choose the inductor ripple current to be approximated 20% - 40% of the maximum input current.

VBUS and PMID Capacitor

The effective capacitance of VBUS and PMID should be enough to absorb the VBUS input switching ripple current. The equation below shows the input capacitor RMS current I_{CIN} calculation.

$$I_{CIN} = \frac{\Delta I}{2 \times \sqrt{3}} \approx 0.29 \times \Delta I \quad (7)$$

Low ESR ceramic capacitors are recommended for input capacitor. The input capacitor voltage ripple can be calculated as follow:

$$\Delta V_{IN} = \frac{V_{IN}}{8 \times f_{SW}^2 \times L \times C_{IN}} \times \left(1 - \frac{V_{IN}}{V_{OUT}} \right) \quad (8)$$

VSYS Capacitor

The VSYS output current of the Boost converter is discontinuous and therefore requires an output capacitor (C_{SYS}) to supply AC current to the load. Ripple current rating of VSYS output capacitor should be higher than the maximum output ripple. The output capacitor RMS current can be calculated by below equation and the maximum value occurs at the highest V_{SYS} and the lowest V_{VBUS} .

$$I_{COUT} = I_{OUT} \times \sqrt{\frac{D}{1-D}} = (I_{CHG} + I_{SYS}) \times \sqrt{\frac{V_{SYS} - V_{VBUS}}{V_{VBUS}}} \quad (9)$$

For the best performance, low ESR ceramic capacitors are recommended. The output voltage ripple can be estimated as follow:

$$\Delta V_{SYS} = \frac{I_{OUT} \times D}{f_{SW} \times C_{SYS}} \quad (10)$$

Layout Guidelines

The switching node (SW) creates very high frequency noises, which are several times higher than f_{SW} (1.5MHz) due to sharp rise and fall times of the voltage and current in the switches. To reduce the ringing issues and noise generation, it is important to design a proper layout for minimizing the current path impedance and loop area. The following considerations can help to make a better layout.

1. Place all the output capacitors as close as possible to SYS and BAT pins. The capacitors ground pins need to be connected to the IC ground with GND plane or short copper trace connections.
2. Place the input capacitor between PMID and GND pins as close as possible to the chip with the shortest copper connections (avoid vias). Choose the smallest capacitor size.
3. Connect one pin of the inductor as close as possible to the SW pin of the device and minimize the copper area connected to the SW node to reduce capacitive coupling from SW area to nearby signal traces. This decreases the noise induced through parasitic stray capacitances and displacement currents to other conductors. SW connection should be wide enough to carry the charging current. Keep other signals and traces away from SW if possible.

APPLICATION INFORMATION (continued)

4. Place output capacitor GND pin as close as possible to the GND pin of the device and the GND pin of input capacitor C_{IN} . It is better to avoid using vias for these connections and keep the high frequency current paths short enough and on the same layer. A GND copper layer under the component layer helps to reduce noise emissions. Pay attention to the DC current and AC current paths in the layout and keep them short and decoupled as much as possible.

5. For analog signals, it is better to use a separate analog ground (AGND) branched only at one point from GND pin. To avoid high current flow through the AGND path, it should be connected to GND only at one point (preferably the GND pin).

6. Place decoupling capacitors close to the IC pins with the shortest possible copper connections.

7. Solder the exposed thermal pad of the package to the PCB ground planes. Ensure that there are enough thermal vias directly under the IC, connecting to the ground plane on the other layers for better heat dissipation and cooling of the device.

8. Select proper sizes for the vias and ensure enough copper is available to carry the current for the given current path. Vias usually have some considerable parasitic inductance and resistance.

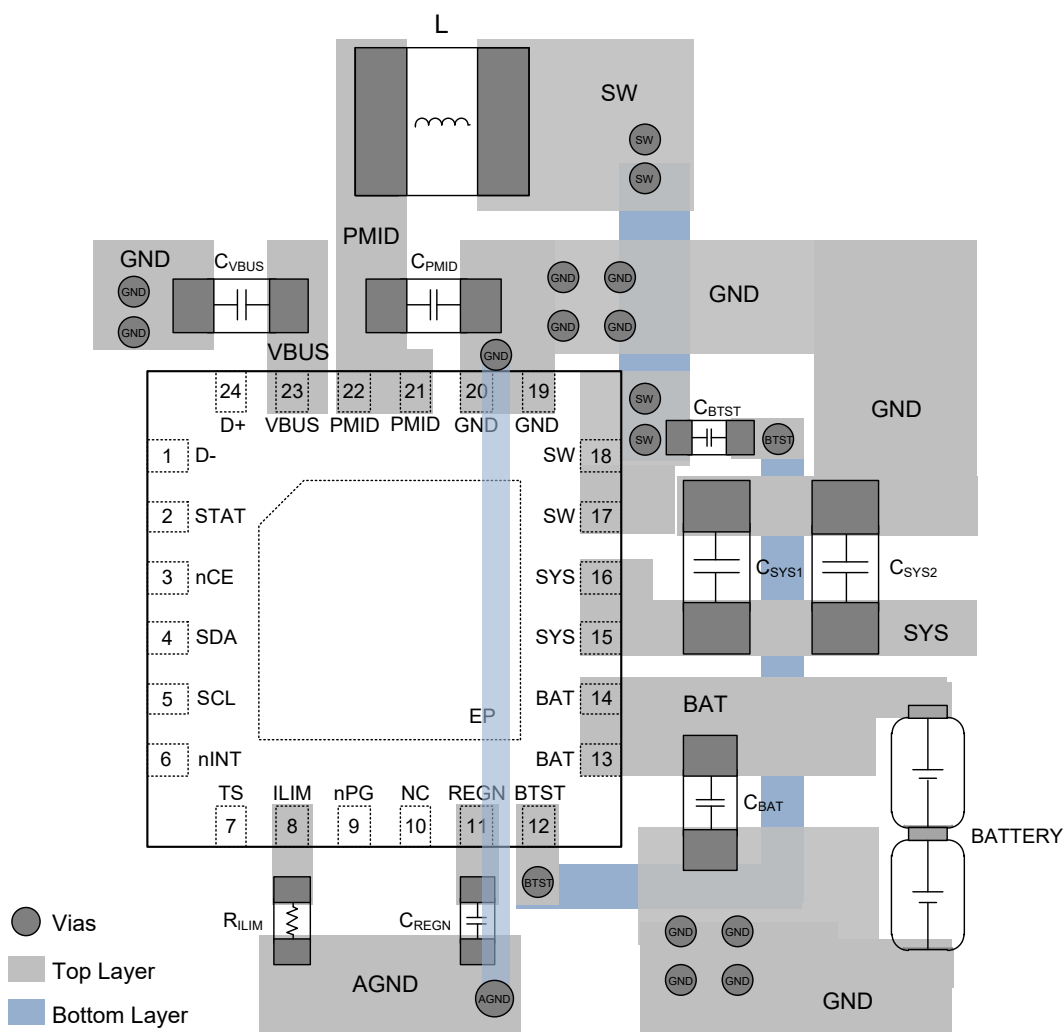


Figure 22. Layout Example

REVISION HISTORY

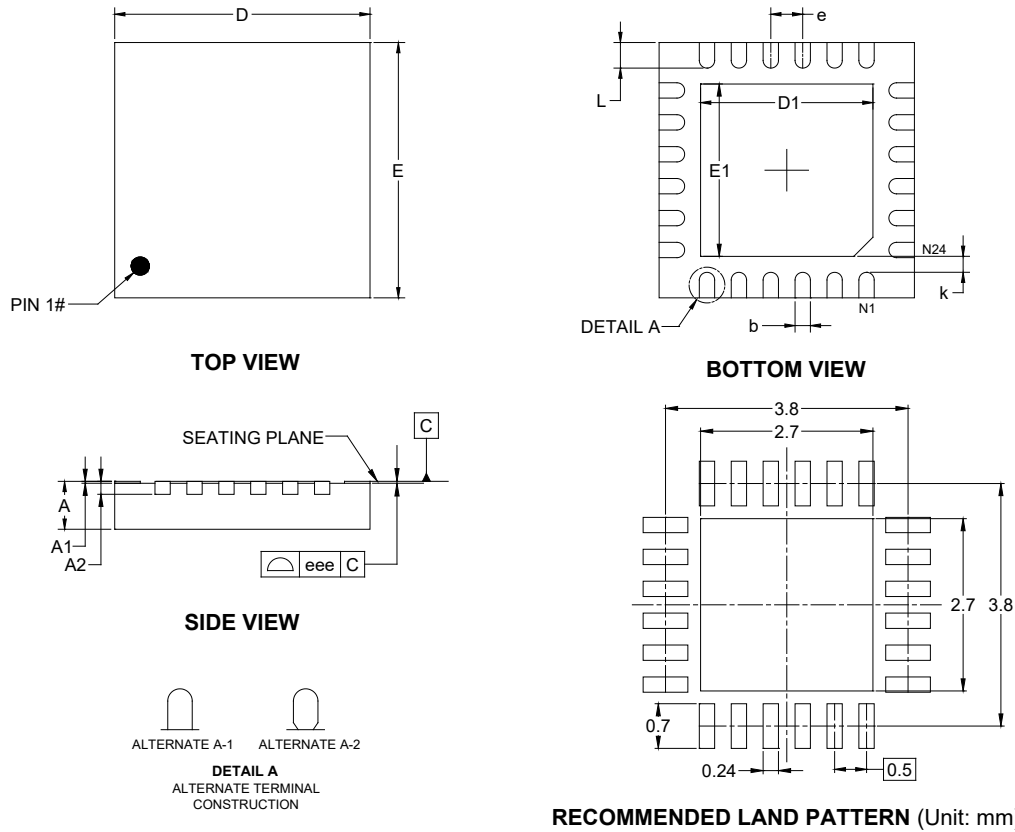
NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

APRIL 2025 – REV.A to REV.A.1	Page
Updated Electrical Characteristics section	8
Updated Detailed Description section	22, 32
Updated Register Maps section	40
Changes from Original (NOVEMBER 2024) to REV.A	Page
Changed from product preview to production data	All

PACKAGE INFORMATION

PACKAGE OUTLINE DIMENSIONS

TQFN-4×4-24L



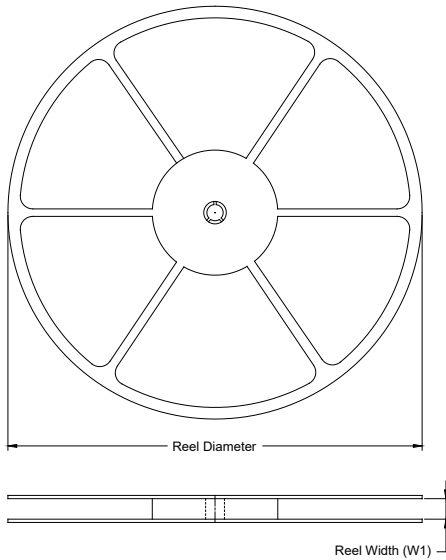
Symbol	Dimensions In Millimeters		
	MIN	NOM	MAX
A	0.700	-	0.800
A1	0.000	-	0.050
A2	0.203 REF		
b	0.180	-	0.300
D	3.900	-	4.100
E	3.900	-	4.100
D1	2.600	-	2.800
E1	2.600	-	2.800
e	0.500 BSC		
k	0.200 MIN		
L	0.300	-	0.500
eee	0.080		

NOTE: This drawing is subject to change without notice.

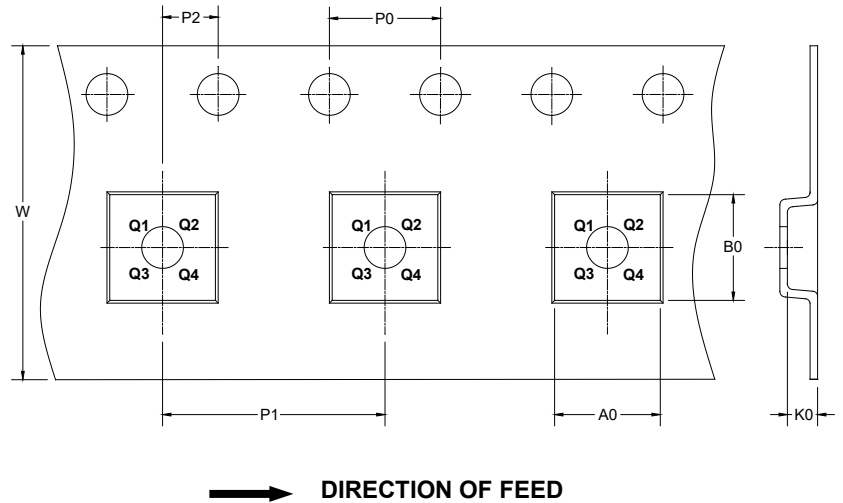
PACKAGE INFORMATION

TAPE AND REEL INFORMATION

REEL DIMENSIONS



TAPE DIMENSIONS



NOTE: The picture is only for reference. Please make the object as the standard.

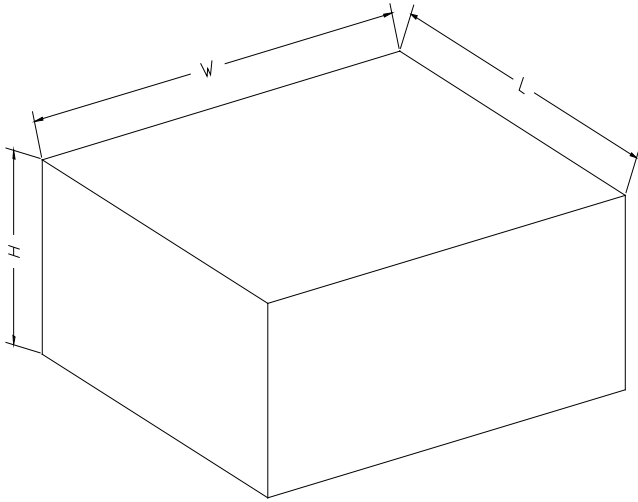
KEY PARAMETER LIST OF TAPE AND REEL

Package Type	Reel Diameter	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P0 (mm)	P1 (mm)	P2 (mm)	W (mm)	Pin1 Quadrant
TQFN-4×4-24L	13"	12.4	4.30	4.30	1.10	4.0	8.0	2.0	12.0	Q2

DD0001

PACKAGE INFORMATION

CARTON BOX DIMENSIONS



NOTE: The picture is only for reference. Please make the object as the standard.

KEY PARAMETER LIST OF CARTON BOX

Reel Type	Length (mm)	Width (mm)	Height (mm)	Pizza/Carton
13"	386	280	370	5

DD0002